

# Theories of deliberated choice

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September 10, 2026

Abstract: To be written.

## 1. Introduction

Acts of choice are the basis of much work in economics. Such acts have been used, among others, to give an observable basis to the concept of preference [Samuelson, 1983].

Most of these approaches consider only acts of choice that are performed spontaneously in the normal course of action of an individual, or that are given a specific frame depending on the psychological aspect that is to be studied. As an example of the former kind, typical willingness to pay studies asks the subject whether she prefers an object at some price or another object at another price, describing the objects in a way that is considered immediately understandable by the subject (or simply showing the objects). Examples of the latter kind include the celebrated work of Kahneman and Tversky [Bell et al., 1988, Kahneman, 2012].

On the other hand, since at least the seminal work of Fishkin [2011], another sort of judgment involving subjective desirability is emerging as also worth studying. These are judgments given by subjects after arguments in favor and against different possible stance have been considered. Such judgments may differ from immediate judgments of desirability given in natural conditions. Indeed, individuals may change opinion on the desirability of some course of action, and therefore revise their choice, while learning about the properties of objects, the consequences of some acts (or empirical knowledge useful to estimate the likelihood of some consequences), or the logical or empirical impossibilities between consequences. This is particularly likely to happen when choosing among non everyday objects or acts whose consequences are not well known by the subject.

This article considers all sorts of influences, including arguments, but more generally texts, images, sounds, experiences (a hiking trip), and so on: anything that can be

transmitted to a subject qualifies as a possible influence. In particular, such an “influence” need not have the logical structure of what is considered a proper argument in argumentation theory. This important feature of this proposal permits to avoid taking a position on what is a correct argument, and permits to study wide enough conceptions of deliberated judgment, including those involving no paternalism [Cailloux and Meinard, 2020]: anything that influences the judgment of a subject, as judged by the subject herself and not by any external standard, may a priori be considered worth including in at least some conceptions of deliberated judgment, thus including whatever some could consider as “fallacies” or “bad reasons”.

While numerous articles have discussed the concept of deliberation in the last decades, how to observe a deliberated judgment in a systematic and reproducible fashion remains unsettled. There is a need to define this concept sufficiently precisely that, first, it is clear what counts as a deliberated judgment (i.e., when “sufficient” [Meinard and Cailloux, 2020] or “correct” deliberation has happened); and second, it is clear which part of the claim is empirical and which part stems from an unfalsifiable conception of what “deliberated” means. This permits to study deliberated judgments empirically, meaning here that disagreements about the deliberated judgment of a set of individuals in a given context can be solved, at least in principle, by an empirical experiment, provided the very conception of what constitutes a deliberated judgment is shared. This article is interested in defining the concept of deliberated judgement of a subject and show how one can study it in an empirical manner. The “classical” counterpart of deliberated judgments, that does not involve confrontation with influences, will henceforth be called the natural judgment. As we deal with choice acts, we will also name the concept “deliberated choice”, as opposed to “natural choice”.

While the concept of deliberation in the deliberative democracy literature usually refers to multiple individuals debating, this article considers deliberation as related to a single individual: an individual deliberates (in the sense of: carefully weights) the stance she considers most appropriate while being confronted to influences. The deliberated judgment of an individual is thus defined with no requirement that the influences be uttered by peers in a debate; they can be given by an automaton, for example. In this sense, deliberated judgments come close to what Rawls [1999] calls reflective equilibrium (referring to an idea of Goodman [1983]), a concept that has much inspired the present proposal.

To define deliberated judgments, one needs to start with the set of influences under consideration, and the way the subject gets to know them, called hereafter the “exposure protocol”. As a result, we need always talk about the deliberated judgment *given some exposure protocol*. The exposure protocol defines a set of influence and, importantly, how influences are sent to the subject and her subsequent choice observed. There is no explicit dependency on the internal process that occurs in the subject’s mind while she (possibly) processes the influences, as it is not desirable to assume that the internal process could be observed. We consider subjects as black boxes and only require a capacity of observation of their choices after exposure to influences. Depending on the exposure protocol and the subject, it is of course possible for the influences to never affect any choice of the subject, a conclusion that may be interesting in its own right.

The deliberated judgment of a subject is defined as a function of the exposure protocol, as accepting the option that resists every counter-influences. The notion of resistance is itself defined according to the exposure protocol.

Studying judgments resulting from exposure to influences includes a difficulty that is not present with natural choice experiments: exposing an individual to influences may change the individual's stance, an effect that cannot be erased in order to submit the individual to an unrelated sequence of influences. The observer thus cannot in general assume that multiple unrelated sequences of influences can be tested on a given individual. This difficulty will be worked around by considering general theories of deliberated judgments, that apply to sets of individuals. This is methodologically similar to defining theories of how objects break. One can break a given (unrepairable) object only once, so such theories must apply to sets of objects if one wants to be able to test multiple ways of falsifying the theories.

For conceptual clarity, I will focus on the case of a single binary decision to take, akin to a choice situation facing a bundle of two courses of action, exactly one of which must be adopted. I will furthermore use a common trick of mathematics by considering sets as infinite to mean that they are sufficiently large that they will not be practically exhausted. More precisely, the set of influences and individuals are considered infinite, not out of some belief that influences are infinitely numerous (I believe that such consideration may occasionally be useful in some philosophical sense but unrealistic in many practical applications), but as a simple modelization of the fact that they will, in most interesting applications, be too numerous to be exhaustively explored: even considering only textual arguments generally considered “relevant”, for many complex topics, it is not reasonable to assume that we can expose an individual to all such arguments before knowing her deliberated judgment. Instead, we need some way to phrase general theories and validate them, in order to reduce the too numerous to the manageable. We model this by the reduction of infinite sets to finite ones.

## 2. Exposure protocol and deliberated choice

I assume given two antagonistic acts, constituting the decision at stake, such as “buy this object for the price of \$15” and “do not buy this object for the price of \$15”, or “take this job offer” and “decline this job offer”. Let the set  $P = \{\varphi, \varphi', 0\}$  denote the set of possible preferences, where  $\varphi$  denote a strict preference for the first act against the other one,  $\varphi'$  conversely, and 0 denotes indifference (both acts are considered equally worth).

### 2.1. Exposure protocol

An exposure protocol is a tuple  $(I, \mathcal{A}, \rightsquigarrow)$ , where  $I$  is a set of individuals,  $\mathcal{A}$  is a set of influences, and  $\rightsquigarrow$ , “leads to”, represents the behavior of each individual as a function of the influences they have been exposed to.

While  $I$  and  $\mathcal{A}$  are fundamental sets whose elements are considered elementary, a few

notations and concepts are required to define  $\rightsquigarrow$ . Throughout the article,  $\mathbb{N}$  designates the natural numbers (including 0) and  $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$  designates the positive numbers. Given  $j, l \in \mathbb{Z}$ , the notation  $\llbracket j, l \rrbracket$  represents  $\{k \in \mathbb{Z} \mid j \leq k \leq l\}$  (thus  $\forall j \in \mathbb{N}^* : \llbracket j, 0 \rrbracket = \emptyset$ ). Define  $\mathcal{A}^{<\mathbb{N}} = \bigcup_{k \in \mathbb{N}} \mathcal{A}^{\llbracket 1, k \rrbracket}$  as the finite sequences of influences, including the empty sequence. A generic element of  $\mathcal{A}^{<\mathbb{N}}$  (a possibly empty finite sequence of influences) is denoted by  $\alpha$  and called an influence path, or simply a path. Its length is denoted by  $\#\alpha \in \mathbb{N}$  and its elements,  $\alpha(k)$  or  $\alpha_k$ . Thus, given an influence path  $\alpha \in \mathcal{A}^{<\mathbb{N}}$ ,  $\forall k \in \llbracket 1, \#\alpha \rrbracket : \alpha(k) = \alpha_k \in \mathcal{A}$ , and  $\alpha = \emptyset \Leftrightarrow \#\alpha = 0$ . Throughout the text, functions are set-theoretically considered as sets of pairs (identified with their graph), so that for example a path is seen as a subset of  $\mathbb{N} \times \mathcal{A}$ . A path  $\alpha \in \mathcal{A}^{<\mathbb{N}}$  is a prefix of a path  $\alpha' \in \mathcal{A}^{<\mathbb{N}}$  whenever  $\alpha \subseteq \alpha'$ , thus,  $\forall k \in \llbracket 1, \#\alpha \rrbracket : \alpha_k = \alpha'_k$ . Given any  $\alpha \in \mathcal{A}^{<\mathbb{N}}$ ,  $\llbracket \alpha \rrbracket = \{\alpha_{\llbracket 1, k \rrbracket} \mid k \in \llbracket 0, \#\alpha \rrbracket\}$  denotes the set of prefixes of  $\alpha$ .

The function  $\rightsquigarrow : I \times \mathcal{A}^{<\mathbb{N}} \rightarrow P$  associates to each individual and path an element of  $P$ ,  $\rightsquigarrow(i, \alpha)$  representing the choice of  $i$  following exposure to the influences of  $\alpha$ , in order. That is,  $\rightsquigarrow(i, \alpha) = \varphi$  denotes that  $i$  chooses the first act after exposure to  $\alpha$ ,  $\rightsquigarrow(i, \alpha) = \varphi'$  denotes  $i$  choosing the second act and  $\rightsquigarrow(i, \alpha) = 0$  is used to represent indifference of  $i$ . The notation  $\alpha \rightsquigarrow_i p$ , pronounced “ $\alpha$  leads  $i$  to  $p$ ”, will be used to mean  $\rightsquigarrow(i, \alpha) = p$ , considering  $\rightsquigarrow_i \subseteq \mathcal{A}^{<\mathbb{N}} \times P$  as a binary relation. As usual with binary relations,  $\alpha \not\rightsquigarrow_i p$  means  $\rightsquigarrow(i, \alpha) \neq p$  or equivalently  $\rightsquigarrow(i, \alpha) \in P \setminus \{p\}$ .

We will also have to consider generic worlds. A world  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$  represents the a priori possible behaviors of individuals in  $I$ . All these worlds but one,  $\rightsquigarrow$ , are counterfactual.

## 2.2. Deliberated choice

The deliberated choice of  $i$  given the exposure protocol  $(I, \mathcal{A}, \rightsquigarrow)$  is denoted  $P_i \subseteq P$ . Defining a deliberated judgment requires a notion of convergence, or at least a kind of stability in our discrete setting, amid this infinite variety of influence paths. This proposal uses the notion of decisive influence, which captures the informal intuition that a judgment is reasonably called deliberate if it rests on some influence that leads to a systematic judgment facing every possible counter-influence.

Given  $\alpha \in \mathcal{A}^{<\mathbb{N}}$ ,  $R(\alpha) = \alpha(\llbracket 1, \#\alpha \rrbracket)$  denotes the range of  $\alpha$ , that is, all influences contained in the sequence  $\alpha$ . Let  $\alpha_{end} = \alpha(\llbracket \max(1, \#\alpha - 1), \#\alpha \rrbracket)$  denote the set containing the last two influences of the sequence  $\alpha$  if the sequence has at least two elements; the unique influence of the sequence if it has exactly one element; and the empty set if it is empty.

Given  $i \in I, p \in P, a \in \mathcal{A}$ , define  $a \hookrightarrow_i p$ , pronounced “ $a$  leads definitely  $i$  to  $p$ ”, iff  $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a \in \alpha_{end} : \alpha \rightsquigarrow_i p$ . When  $a \hookrightarrow_i p$ ,  $a$  is said to be a decisive influence for  $p$  from  $i$ ’s point of view given the exposure protocol. Another interpretation of this definition is that  $a$  then resists every counter-influences, in the sense that no influence  $a'$  given at the end of any sequence of influences, in presence of  $a$ , leads the individual to  $p' \neq p$  (formally,  $\nexists a' \in \mathcal{A}, \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a, a' \in \alpha_{end} \wedge \alpha \rightsquigarrow_i p'$ ).

The deliberated choice of  $i$  is then considered as the set of propositions supported by decisive influences.

**Definition 1** (Deliberated choice).  $\forall i \in I, p \in P : p \in P_i \Leftrightarrow [\exists a \in \mathcal{A} \mid a \hookrightarrow_i p]$ .

It follows from the consideration of “end” as the last *two* influences that the whole set of decisive influences can support at most a single proposition, thus, that the deliberated choice cannot hold more than one proposition.

**Theorem 1** (Deliberated choice singularity).  $\forall i \in I : \#P_i \leq 1$ .

*Proof.* Assume that  $\exists a, a' \in \mathcal{A}, p, p' \in P \mid a \hookrightarrow_i p \wedge a' \hookrightarrow_i p'$ . Since  $a \hookrightarrow_i p$ ,  $(a, a') \rightsquigarrow_i p$ , and since  $a' \hookrightarrow_i p'$ ,  $(a, a') \rightsquigarrow_i p'$ , hence  $p' = p$ .  $\square$

The intended semantics is that  $P_i$  represents the preference of  $i$  over the proposed set of two acts, after deliberation, given the exposure protocol. In other words, if  $P_i = \{\varphi\}$ , the individual considers the first act strictly better than the second one, after deliberation (and conversely for  $P_i = \{\varphi'\}$ ); if  $P_i = \{0\}$ , the deliberated judgment of  $i$  is that both acts are equally worth; and if  $P_i = \emptyset$ , no act is deliberately strictly better than the other one and they are also not deliberately exactly equivalent to each other.

This latter possibility may look unpleasant, but it is a fact that facing some difficult problems, individuals sometimes are undecided, and it is an a priori empirical possibility that the individual remains undecided (or even, *becomes* undecided) when considering multiple influences. Also, being undecided is not the same state as being decidedly indifferent about two issues.

Given a world  $\rightsquigarrow$ , deliberated choices are represented by an element  $\pi(\rightsquigarrow) \subseteq I \times P$  defined as  $\pi_i(\rightsquigarrow) = \pi(\rightsquigarrow)(i) = P_i$ . Theorem 1 shows that  $\pi(\rightsquigarrow)$  is right unique (meaning that  $\forall i \in I : \#\pi(\rightsquigarrow)(i) \leq 1$ ). Equivalently, note that  $(P \cup \{\emptyset\})^I$ , where  $\{\emptyset\}$  is used as a sentinel value for individuals with no deliberated judgment, can be interpreted as the right unique relations over  $I \times P$  (discarding those individuals with no deliberated judgment); and then the deliberated choices are an element  $\pi(\rightsquigarrow) \in (P \cup \{\emptyset\})^I$ .

The deliberated judgment of a subject depends on the exposure protocol, that is, on the set of influences and the way they are submitted to the subject, and not only on the set of influences, because the very way that they are transmitted to the individual may affect her judgment differently. For example, reading an image description and showing an image to a subject may impress them differently [Railton, 2003].

### 2.3. An example and a discussion about observability

The following toy example is inspired by the following argument from Railton [2003, p. 59].

An experiment in cognitive psychology reveals that when subjects observe two individuals taking a test, one of whom answers a high percentage of the early questions correctly but then falters, while the other does poorly initially but then answers a high percentage of the later questions correctly, the first is usually viewed as more able, even though in the end each answers the same number of questions correctly in all. If the subjects had been asked which test-taker they would prefer as a math tutor, they likely would have answered

“The first,” and perhaps they would also have been willing to pay a premium to secure his services. But if they were to become convinced of the influence of sheer order on their evaluations of relative ability, they presumably would no longer want their initial preference for the first test-taker to influence their choices – or willingness to pay – in this way.

*Example 1* (Math tutor). Let  $I$  be a set of individuals desiring to evaluate two candidates  $\{m_1, m_2\}$  for math tutoring. Consider one individual  $i \in I$ . Let  $\rightsquigarrow(i, a_1)$  represent  $i$  being shown a test of candidate  $m_1$  wherein  $m_1$  answers 70% of the first half of questions correctly but then only 40% of the other half correctly. Let  $\rightsquigarrow(i, a_2)$  represent  $i$  being shown a test of candidate  $m_2$  wherein  $m_2$  answers 45% of the first half of questions correctly, then 75% of the other half correctly. Let  $\rightsquigarrow(i, a_e)$  represent  $i$  being given the following influence. “When tested,  $m_1$  answers 70% of the first half of questions correctly but then only 40% of the other half correctly; whereas  $m_2$  answers 45% of the first half of questions correctly, then 75% of the other half correctly. It follows that  $m_1$  answers on average less well than  $m_2$ . Observing the candidates take the tests may yield a contrary impression; this is a well-known psychological phenomenon depending on the order of the questions that the candidates answer correctly.” Let  $\varphi_1$  represent choosing  $m_1$  and  $\varphi_2$  choosing  $m_2$  (for this example we use  $P = \{\varphi_1, \varphi_2, 0\}$ ).

If the prediction described in the quote applies to  $i$ , the relation  $\rightsquigarrow_i$  satisfies,  $\forall k \in \mathbb{N}^*$  (recall that for  $A \subseteq \mathcal{A}$ ,  $A^k$  represents the set of sequences of length  $k$  using influences from  $A$ ):  $\forall \alpha \in \{a_1, a_2\}^k, \alpha \rightsquigarrow_i \varphi_1$  and  $\forall \alpha \in \mathcal{A}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$ . The latter in particular implies:  $\forall \alpha \in \{a_e\}^k, \alpha \rightsquigarrow_i \varphi_2$ ;  $\forall \alpha \in \{a_1, a_e\}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$ ;  $\forall \alpha \in \{a_2, a_e\}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$ . The values of  $\rightsquigarrow(i, \alpha)$  for  $\alpha = \emptyset$  and for  $\alpha \in \mathcal{A}^k \mid a_e \notin \alpha_{end}$  are undetermined by the prediction, but whatever they are,  $a_e$  is a decisive influence for  $\varphi_2$ , hence,  $i$ ’s deliberated choice is  $\varphi_2$ .  $\triangle$

Now that deliberated choices are defined, we need to explore what one can learn about deliberated choices using our empirically available data. This requires to define claims about deliberated choices and to explore how to validate claims.

### 3. Claims about deliberated choice

Given an exposure protocol  $(I, \mathcal{A}, \rightsquigarrow)$ , consider a claim about  $\rightsquigarrow$ , such as “every individuals in  $I$  choose deliberately  $\varphi$ ”, or, in example 1, “ $a_e$  is a decisive influence for  $i$  in favor of  $\varphi_2$ ”. Validating claims about deliberated choice is no trivial matter, for two reasons.

First, such claims must entail an infinity of constraints about  $\rightsquigarrow$ : in order to deduce that  $p \in P_i$  for some  $p \in P, i \in I$ , the claim must constrain  $\rightsquigarrow_i$  on every possible paths  $\alpha \in \mathcal{A}^{<\mathbb{N}}$ . It follows that a claim cannot be validated empirically exhaustively by observing the value of  $\rightsquigarrow$  on every points that it constraints.

Second, our observations are limited in important ways: once we have shown some influences to a given individual  $i$ , we cannot undo the effects of this action on  $i$  and show to her a different influence as if starting from scratch (though one can in some

circumstances assume  $i$  forgets influences after a while; indeed we will study precisely which of these assumptions are necessary and sufficient to permit studying deliberated choice). In other words, once we have shown a path  $\alpha$  to  $i$ , we can only continue the protocol with  $i$  by showing paths  $\alpha'$  that extend  $\alpha$ .

To illustrate this point with example 1, let  $\alpha = (a_1, a_e, a_2)$  designate the influence path where the influences are shown in the order they are described in example 1, and let  $\alpha' = (a_2, a_e, a_1)$  designate the reverse order. The deliberated judgment does not depend on the order of the influences shown to  $i$ , as long as the end of the path features a decisive influence: for having  $p \in P_i$ , it must be that directing to  $i$  the path  $\alpha$  or the path  $\alpha'$  both lead  $i$  to choose  $p$ . But it is impossible to observe both the value of  $\rightsquigarrow(i, \alpha)$  and the value of  $\rightsquigarrow(i, \alpha')$ .

The first difficulty is familiar as it occurs in any endeavor to validate general claims in science: claims that cover a large (or infinite) spectrum cannot be validated by observing the system exhaustively. One must rather develop a falsification protocol, that is, a way to invalidate the claim, and then observe that the claim is not invalidated. This is the approach taken in the next section.

This approach also permits to tackle the second difficulty, by letting us study theoretically the needs and limits of falsification given our observable basis.

### 3.1. Claims and observable basis

Note first that given a relation  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ,  $\mathcal{P}(\rightsquigarrow)$  denotes the set of relations over  $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$  that refine  $\rightsquigarrow$  (thus, each element  $f \in \mathcal{P}(\rightsquigarrow)$  is right-unique but non necessarily total, meaning that it does not necessarily cover  $I \times \mathcal{A}^{<\mathbb{N}}$ ). Given sets  $A$  and  $B$ ,  $A \subseteq_{\text{fin}} B$  denotes that  $A$  is a finite subset of  $B$ , thus  $A \subseteq_{\text{fin}} B$  iff  $A \subseteq B$  and  $A$  is finite. Given a function  $f$  on some domain  $\mathcal{A}$ , given  $A \subseteq \mathcal{A}$ , we write  $f(A) = \{f(a) \mid a \in A\}$  to denote the images of each element in  $A$ . This creates no ambiguity because the functions appearing in this article apply on domains whose elements are never subsets of the domain (thus for a domain  $A$ ,  $a \in A \Rightarrow a \not\subseteq A$ ).

Given a pair  $(I, \mathcal{A})$ , a claim  $C$  is a non-empty set of possibilities for  $\rightsquigarrow$ , thus,  $\emptyset \neq C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ . For brevity, claims will often be described using just a formula in brackets, for example  $C = [\forall i, \exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \alpha \rightsquigarrow_i \varphi]$  is the claim  $C = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i, \exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \alpha \rightsquigarrow_i \varphi\}$ . [OC: I sometimes use bold  $\rightsquigarrow$  instead of  $\rightsquigarrow$ , to be fixed.] A claim is trivial iff it is the trivial claim, defined as  $C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ . Let  $C^\subseteq = \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow) = \{c \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid \exists \rightsquigarrow \in C \mid c \in \mathcal{P}(\rightsquigarrow)\}$  denote the set of non necessarily complete possibilities compatible with  $C$  (thus, each element  $c \in C^\subseteq$  is right-unique but non necessarily total, meaning that it does not necessarily cover  $(I \times \mathcal{A}^{<\mathbb{N}})$ ).

Define the increasing sets of paths as the set  $\mathcal{I}(\mathcal{A}^{<\mathbb{N}}) = \{H \subseteq_{\text{fin}} \mathcal{A}^{<\mathbb{N}} \mid \forall \alpha, \alpha' \in H : \#\alpha \leq \#\alpha' \Rightarrow \alpha \subseteq \alpha'\} \subseteq \mathcal{P}(\mathcal{A}^{<\mathbb{N}})$ . An increasing set of paths  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  can be strictly ordered by the lengths of the paths in  $H$  and can thus also be considered as a sequence of paths. Note that  $\mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  includes both the empty set of paths  $\emptyset$  and the set of paths  $\{\emptyset\}$  containing only the empty path. A non empty set of paths  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  has a unique maximal element  $\{\max H\} = \{\alpha \in H \mid \forall \alpha' \in H : \alpha' \subseteq \alpha\}$ . A non empty set of paths  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  contains all its prefixes (meaning,  $\forall \alpha \in H : \{\alpha' \subseteq \alpha\} \subseteq H$ ).

iff  $H = \llbracket \max H \rrbracket$ . Let  $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$  denote the functions  $f : I \mapsto \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  with finite support, thus, satisfying  $\#f^{-1}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}) \setminus \emptyset) \in \mathbb{N}$ . An element  $Q \subseteq I \times \mathcal{A}^{<\mathbb{N}}$  is called a query iff  $\forall i \in I : Q(i) = \{\alpha \in \mathcal{A}^{<\mathbb{N}} \mid (i, \alpha) \in Q\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  and  $\#Q \in \mathbb{N}$ . Abusing notations, we also view  $Q$  as an element of  $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ . An observation associates to a query an element of  $P$  per queried element, thus, the set of possible observations is  $\Theta = \cup_{Q \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} P^Q$ . In other words, defining a general observation  $O$  as a right unique finite subset of  $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$ , an observation  $\theta \in \Theta$  is a general observation satisfying  $\forall i \in I, (i, \alpha, p), (i, \alpha', q) \in \theta : \# \alpha \leq \# \alpha' \Rightarrow \alpha \subseteq \alpha'$ . Given a general observation  $O$ , let  $O(P) \subseteq I \times \mathcal{A}^{<\mathbb{N}}$  denote the queried elements (the pairs  $(i, \alpha)$ ) in  $O$  and  $O(\mathcal{A}^{<\mathbb{N}}, P)$  designate the observed individuals. Note that given an observation  $\theta$ ,  $\theta(P) \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ . Similarly, we can view an observation  $\theta$  as a function  $\theta : I \mapsto \cup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H$  and thus  $\Theta = \mathcal{F}((\cup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H)^I)$ , where  $\mathcal{F}$  designates the finite relations (considered as a set of triples). An observation  $\theta'$  rightward-extends an observation  $\theta$ , denoted  $\theta \sqsubseteq \theta'$ , iff  $\theta \subseteq \theta' \wedge \forall i \in I, (i, \alpha, p) \in \theta, (i, \alpha', p') \in \theta' : \alpha \subseteq \alpha'$ .

An anonymous observation  $\mathbf{g} \in \cup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H$  associates each path from an increasing set of paths to an element of  $P$ . We let  $\{i\} \times \mathbf{g} = \{(i, \alpha, p) \mid (\alpha, p) \in \mathbf{g}\} \in \Theta$  denote the observation corresponding to applying  $\mathbf{g}$  to  $i$ ; and  $\mathbf{g}^{-1}(P) = \{\alpha \mid \exists p \in P \mid (\alpha, p) \in \mathbf{g}\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  the used paths.

We say that an observation  $\theta$  is compatible with a world  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$  iff  $\theta \subseteq \rightsquigarrow$  (seeing the world as a set of triples). Thus, given  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , the set  $\Theta \cap \mathcal{P}(\rightsquigarrow)$  is the set of observations compatible with  $\rightsquigarrow$ . Similarly, an observation  $\theta$  is compatible with a claim  $C$  iff  $\theta \in \cup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$ , equivalently, iff  $\theta \in C^\subseteq$ . The set  $\Theta \cap C^\subseteq$  is the set of observations compatible with the claim. Given a claim  $C$  and an observation  $\theta$ ,  $\pi(\uparrow \theta \cap C) = \{\pi(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\}$  thus denotes the DJ functions (mappings of  $I$  to  $P \cup \{\emptyset\}$ ) compatible with the observation and the claim.

A subset of triples  $T \subseteq I \times \mathcal{A}^{<\mathbb{N}} \times P$  is unobservable iff it cannot be observed together, that is, iff  $T \notin \Theta$ . If  $\#T \notin \mathbb{N}$ ,  $T$  is unobservable. A finite subset of triples  $T$  is unobservable iff some subset  $T' \subseteq T$  with  $\#T' = 2$  is unobservable (unobservability comes from a given individual's observations, for that one, the paths are not increasing, and as the subset relation is transitive, two elements suffice to determine this fact).

It is important to realize that  $\Theta$  is the set of *a priori* possible observations, meaning, those that are conceivable, possibly counter-factually. The set of observations that can result from actually observing the world  $\rightsquigarrow$  is  $\Theta \cap \mathcal{P}(\rightsquigarrow) = \{\rightsquigarrow|_Q \mid Q \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\}$ . Such an observation  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  is called a feasible observation.

Given a right unique  $\tau \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P$ , we let  $\uparrow \tau = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \tau \subseteq \rightsquigarrow\}$  denote the set of “leads to” relations that complete  $\tau$  (thus  $\uparrow \emptyset = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ).

## 3.2. Requests

Onwards, we assume given an exposure protocol  $(I, \mathcal{A}, \rightsquigarrow)$ .

To be interesting for our purpose, a claim must satisfy minimal desiderata.

Most important, of course, is that the claim includes the real world. A claim  $C$  *holds* iff  $\rightsquigarrow \in C$ . We will study criteria to determine whether a claim holds.

Once established that a claim holds, any feasible observation can be combined with the claim to determine conclusions. We want, minimally, that at least some feasible observation permits to conclude about at least one individual's DJ. Furthermore, we want such observations to be always available even after some previous observations have occurred already, to ensure reproducibility. A claim  $C$  is *conclusive* iff  $\forall \rightsquigarrow \in C, \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta' \sqsubseteq \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), i \in I, p \in P \mid \pi_i(\uparrow \theta' \cap C) = \{p\}$ .

We restrict attention to claims that do not use prior knowledge about named individuals, thus, are anonymous, in the following sense. Given a permutation  $\sigma$  over  $I$  and any  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , let  $\rightsquigarrow \circ \sigma$  denote  $\{(i, \alpha, \varphi) \mid (\sigma(i), \alpha, \varphi) \in \rightsquigarrow\}$  and let  $C \circ \sigma$  denote  $\{\rightsquigarrow \circ \sigma \mid \rightsquigarrow \in C\}$ . A claim  $C$  is *anonymous* iff  $\forall \sigma : C = C \circ \sigma$ . For example, given  $a$ ,  $i$  and  $\varphi$ , the claim  $[(a) \rightsquigarrow_i \varphi]$ , which claims that a given path leads a given individual to a given proposition, is not anonymous (example 5); whereas given  $a$  and  $\varphi$ , the claim  $[\forall i : (a) \rightsquigarrow_i \varphi]$  is anonymous.

### 3.3. Verifiability

We want to know whether a claim holds, but all we have under hand are observations. A natural starting point is to attempt to verify claims, defined as follows.

The claim  $C$  is verified, or satisfies verifiedness, iff some observation compatible with the actual world proves the claim, in the sense that every world compatible with the observation is in  $C$ ; formally, iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \uparrow \theta \subseteq C$ .

Verifiedness is sound (corollary 1), meaning that indeed when a claim is verified, it holds, but is too demanding for many claims, that is, many claims fail the converse implication. Indeed, some sufficiently simple claims, such as the claim  $C = \{\rightsquigarrow \mid (a) \rightsquigarrow_i \varphi\}$  that some given path leads some given individual to some given proposition, do satisfy verifiedness (provided they hold), but such claims are not strong enough to deduce much useful conclusion concerning anyone's DJ.

It turns out that any claim that is conclusive fails verifiedness. This is even visible by considering a weaker property than conclusiveness. Given a claim  $C$  and a world  $\rightsquigarrow \in C$ , the claim is *informative in*  $\rightsquigarrow$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \pi(\uparrow \theta \cap C) \neq (P \cup \{\emptyset\})^I$ . Being informative in  $\rightsquigarrow$ , a weaker demand than conclusiveness, implies failure of verifiedness (corollary 2). This may be expected, as we did not postulate yet any sort of regularity in the world, and indeed the fundamental reason for the incompatibility is that any observation can be completed in a way that contradicts any conclusion that informativeness in a world mandates (lemma 1).

We thus now turn to examining which sort of prior knowledge will suffice for ensuring the existence of a verification procedure that will allow to determine that a conclusive and anonymous claim holds (when it indeed holds). As such prior knowledge cannot come from observations, we call them axioms.

Three sorts of axioms will be involved, the two first of which can be seen as applications to our context of Popperian ideas. First, we will assume that falsification tests fail falsifying the claim, under condition of sufficient observability. Second, we will assume that behaviors are in sufficient supply for being observed without bound. Third, we will

assume that some behaviors that are inobservable (thus impossible to reject empirically) do not exist.

The following example illustrates the tension that needs overcoming. Demanding that a claim be verifiable permits the claim to contain punctual facts, but conclusiveness requires general statements (this informal explanation is used to build intuition and will be made more precise in the course of our reasoning).

*Example 2.* Consider the world  $\rightsquigarrow$  constituted of two kinds of individuals, the jay's which always answer  $\varphi$  and the kay's which always answer  $\varphi'$ . Thus,  $I = J \cup K$ , and  $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}} : [i \in J \Rightarrow \alpha \rightsquigarrow_i \varphi] \wedge [i \in K \Rightarrow \alpha \rightsquigarrow_i \varphi']$ . The claim  $C_1 = [\exists i \mid \emptyset \rightsquigarrow_i \varphi]$  is verifiable and not conclusive. The claim  $C_2 = [(\forall i : \emptyset \rightsquigarrow_i \varphi \Rightarrow \forall \alpha, \alpha \rightsquigarrow_i \varphi) \wedge (\exists i \mid \emptyset \rightsquigarrow_i \varphi)]$  is conclusive and not verifiable.  $\triangle$

We will see that combining verification procedures, which permit to empirically verify punctual facts, with axioms that deal with generalities, lead further than procedures based only on axioms or on verification could.

We say that an axiom  $D$  proves a claim  $C$  iff  $D \subseteq C$ .

### 3.4. Verifiability under an axiom

Verifiability, as defined in section 3.3, mandates that some observation proves  $C$ . As indicated there, this is too demanding to be applicable with no prior knowledge. In this section, we extend this idea to the case where some axiom is given.

$C$  is verified at world  $\rightsquigarrow$  under axiom  $D$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \emptyset \neq \uparrow \theta \cap D \subseteq C$ .  $C$  is verified under  $D$  iff it is verified at  $\rightsquigarrow$  under  $D$ . Note that  $C$  is verified at  $\rightsquigarrow$  iff it is verified at  $\rightsquigarrow$  under the trivial axiom.  $C$  satisfies established verifiability under axiom  $D$  iff it is verified under  $D$  at all worlds in  $C \cap D$ , thus:  $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \uparrow \theta \cap D \subseteq C$ .

Theorem 2 shows that provided the axiom is true, being verified under an axiom suffices for holding.

Provability by  $D$  implies established verifiability under  $D$  (just use verification  $\theta = \emptyset$ ). For some axioms, provability by  $D$  is strictly stronger than established verifiability under  $D$  (the claim  $C = [(a) \rightsquigarrow_i \varphi]$  satisfies established verifiability under the trivial axiom  $D$  but is not provable by it).

Established verifiability only requires *some* proving observation to exist at each  $\rightsquigarrow \in C \cap D$ . It is desirable that every prefix admit a rightward extension that still proves  $C$ . Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ,  $C$  is permanently verifiable under axiom  $D$  iff  $\forall \rightsquigarrow \in C \cap D, \forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta' \sqsubseteq \theta \mid \uparrow \theta' \cap D \subseteq C$ . Under a given axiom, permanent verifiability is stronger than established verifiability (corollary 6).

## 4. Priors

### 4.1. Falsifiability

Our first axiom assumes that all falsification tests against  $C$  fail.

Expressing this requires properties that describe the extend to which a given (hypothetical or real) world can be observed to be not the real world. Given  $\rightsquigarrow$  and  $C$ , say that  $C$  is

False in  $\rightsquigarrow$  ( $F_C(\rightsquigarrow)$ ) iff  $\rightsquigarrow \notin C$ ;

Observably False (OF) in  $\rightsquigarrow$  ( $OF_C(\rightsquigarrow)$ ) iff  $C$  is false in this world and its falsity is observable; formally, iff  $[\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq]$ , equivalently,  $\Theta \cap \mathcal{P}(\rightsquigarrow) \not\subseteq C^\subseteq$ ;

Persistently Observably False in  $\rightsquigarrow$  ( $POF_C(\rightsquigarrow)$ ) iff  $C$  is false in this world and its falsity is observable whatever past observation have occurred already; formally, iff  $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta' \notin C^\subseteq$ .

We sometimes write  $POF_C$  instead of  $POF_C(\rightsquigarrow)$  when the world  $\rightsquigarrow$  is clear from the context, and  $POF$  when  $C$  is clear from the context (or referring to any generic claim), and similarly for  $OF_C$  and  $F_C$ .

Permanent verifiability and  $POF$  have a dual character as  $C$  is permanently verifiable under  $D$  iff  $\forall \rightsquigarrow \in C \cap D, POF_{D \setminus C}(\rightsquigarrow)$  (corollary 5, see also remark 9).

Note that  $\forall C, \rightsquigarrow : POF_C(\rightsquigarrow) \Rightarrow OF_C(\rightsquigarrow) \Rightarrow F_C(\rightsquigarrow)$  (theorem 6 proves the second implication). Also, given  $C' \subseteq C$ ,  $\forall \rightsquigarrow : \neg F_{C'}(\rightsquigarrow) \Rightarrow \neg F_C(\rightsquigarrow)$ ,  $\neg OF_{C'}(\rightsquigarrow) \Rightarrow \neg OF_C(\rightsquigarrow)$ , and  $\neg POF_{C'}(\rightsquigarrow) \Rightarrow \neg POF_C(\rightsquigarrow)$ .  $C$  is said to suffer visibility issues in  $\rightsquigarrow$  iff  $\neg POF_C(\rightsquigarrow) \wedge OF_C(\rightsquigarrow)$ , which happens when some past observation (called “prefix” in this context) hides the falsity of  $C$  even though its falsity is observable when no prefix is involved.  $C$  suffers visibility issues iff  $\exists \rightsquigarrow \mid \neg POF_C(\rightsquigarrow) \wedge OF_C(\rightsquigarrow)$ .  $C$  suffers no visibility issues in  $\rightsquigarrow$  iff  $OF_C(\rightsquigarrow) \Rightarrow POF_C(\rightsquigarrow)$ , equivalently, iff  $OF_C(\rightsquigarrow) \Leftrightarrow POF_C(\rightsquigarrow)$ , and suffers no visibility issues iff  $\forall \rightsquigarrow : OF_C(\rightsquigarrow) \Rightarrow POF_C(\rightsquigarrow)$ .

We say that  $C$  is *falsifiable* iff  $C \neq P^{I \times \aleph^{< \mathbb{N}}} \wedge [F_C(\rightsquigarrow) \Rightarrow OF_C(\rightsquigarrow)]$ .

It is important to note that falsifiability does not depend on  $\rightsquigarrow$ : whether falsifiability holds can be determined analytically, meaning, by considering only the content of the claim, without considering the world.

We introduce the axiom called  $C$ -scrutiny, which postulates that the real world is such that  $C$  is either not observably false in it, or if it is, then  $C$  suffers visibility issues in it:

**Axiom 1** ( $C$ -scrutiny).  $\neg OF_C(\rightsquigarrow) \vee (\neg POF_C(\rightsquigarrow) \wedge OF_C(\rightsquigarrow))$

This is logically equivalent to mandating  $\neg POF_C(\rightsquigarrow)$ , thus, excluding worlds in which  $C$  is persistently observably false.

*Remark 1.* Another way to describe  $C$ -scrutiny is that if the world is observably out of  $C$ , there is some prefix observation that hides this fact (formally:  $OF_C(\rightsquigarrow) \Rightarrow OF_C(\rightsquigarrow) \wedge \neg POF_C(\rightsquigarrow)$ ).

Equivalently, if  $C$  is still observably falsifiable after any amount of observation, then no feasible observation would falsify it ( $POF_C(\rightsquigarrow) \Rightarrow \neg OF_C(\rightsquigarrow)$ ).

Assuming, as the axiom does,  $\neg POF_C(\rightsquigarrow)$ , leaves three possibilities: either there are visibility issues ( $OF_C(\rightsquigarrow) \wedge \neg POF_C(\rightsquigarrow)$ ), or the claim is false, but inobservably so ( $F_C(\rightsquigarrow) \wedge \neg OF_C(\rightsquigarrow)$ ), or the claim holds ( $\neg F_C(\rightsquigarrow)$ ).  $\triangle$

*Remark 2.* When considering a claim  $C$  with no visibility issues,  $C$ -scrutiny is equivalent to  $\neg \text{OF}_C(\rightsquigarrow)$ .

The reason for adopting the axiom  $\neg \text{POF}_C(\rightsquigarrow)$  instead of  $\neg \text{OF}_C(\rightsquigarrow)$  is illustrated with  $C_1$  is  $(a) \rightsquigarrow_i \varphi \wedge (a') \rightsquigarrow_i \varphi$  for a given  $i$ : in some world (for example, where  $(a) \rightsquigarrow_i \varphi$  even though  $(a') \not\rightsquigarrow_i \varphi$ ),  $\neg \text{POF}_{C_1}(\rightsquigarrow) \wedge \text{OF}_{C_1}(\rightsquigarrow)$ , thus we cannot assume that non-observation of falsification implies not OF, because it might also be due to observing that  $(a) \rightsquigarrow_i \varphi$ . It also applies to  $C_2 = [\forall i : (a') \rightsquigarrow_i \varphi \wedge (a) \rightsquigarrow_j \varphi]$  for a given  $j$ , which in a world satisfying  $(a) \not\rightsquigarrow_j \varphi$  and  $\forall i : (a') \rightsquigarrow_i \varphi$  is  $\neg \text{POF}_{C_2}(\rightsquigarrow)$  (consider the prefix  $\theta = \{(a') \rightsquigarrow_j \varphi\}$  and  $\text{OF}_{C_2}(\rightsquigarrow)$ . (Note that if  $C_2$  fails because, for some  $i$ ,  $(a') \not\rightsquigarrow_i \varphi$ , then  $\theta$  is not a hiding prefix observation, conforming to intuition.)  $\triangle$

Remark 8 discusses alternative formalizations of the concepts POF and  $C$ -scrutiny.

Theorem 19 shows that this axiom is insufficient: only the trivial claim is permanently verifiable under  $\neg \text{POF}_C$ .

## 4.2. Repeatable observations

Our second axiom assumes that whenever a behavior is exhibited by at least one individual, infinitely many individuals exhibit it.

**Axiom 2** (Repeatable observations). *Let  $E = [\forall i \in I : \#\{j \in I \mid \rightsquigarrow_j = \rightsquigarrow_i\} \notin \mathbb{N}]$ , the claim that every individual's behavior is shared by infinitely many others.*

Given  $C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , combining both axioms introduced so far, we obtain the axiom  $F_C = (\neg \text{POF}_C) \cap E$ .

Intuition might suggest that under  $E$ , no visibility issues can arise. Example 15 shows that, on the contrary, a claim can be provable by  $\neg \text{OF}_{C \cap E} \cap E$  yet not provable by  $\neg \text{POF}_{C \cap E} \cap E$ .

Theorem 22 shows that  $(\neg \text{POF}_C) \cap E$  proves  $C$  iff  $(\neg \text{OF}_C) \cap E$  proves  $C$  and  $C$  suffers no visibility issues in  $E$  (that is,  $\forall \rightsquigarrow \in E : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$ ).

Theorem 24 shows that anonymous claims suffer no visibility issues in  $E$ . Thus (using theorem 22), any anonymous and falsifiable claim  $C \subseteq E$  is provable by  $\neg \text{POF}_{C \cap E} \cap E$  (theorem 25).

For any anonymous and conclusive claim to exist that holds and is permanently verifiable under  $\neg \text{POF}_{C \cap E} \cap E$ , we need the world to offer some systematic pattern of the form, whenever an individual satisfies some observable behavior, then she deliberately chooses a given proposition (theorem 32). Assume that the world has such:  $\exists \mathbf{g}, i, a, p \mid \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow \exists p \in \pi(\rightsquigarrow_i)$ . We want to study axioms for which, for worlds which satisfy such conditions, some anonymous, conclusive, holding claim exists that is permanently verifiable under that axiom.

Theorem 30 shows the insufficiency of the axiom  $F_C$ : for “most” such worlds (in some technical sense to be fully elucidated), there exists no anonymous claim that holds and is conclusive and permanently verifiable under  $\neg \text{POF}_{C \cap E} \cap E$ .

### 4.3. Visibility

We now add the last piece of the puzzle. For a suitable claim to exist, we need the axiom to ensure that no behavior is sufficiently “strange” that an individual  $i$  can satisfy the pattern  $\mathbf{g} \subseteq \rightsquigarrow_i$ , but fail  $a \hookrightarrow_i p$ , but only fail it in a way that is invisible to our observational capacities. In words, we request that those individuals have some path compatible with those used in  $\mathbf{g}$  where it is visible that  $a$  does not decisively lead  $i$  to  $p$  (thus  $a$  appears at the end of  $\alpha$  and yet  $\alpha \not\rightsquigarrow_i p$ ). In formal terms, we impose  $D^{\mathbf{g},a,p}$ , defined as follows.

**Axiom 3** (Visibility).

$$\forall i : (\mathbf{g} \subseteq \rightsquigarrow_i \wedge a \not\rightsquigarrow_i p) \Rightarrow (\exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \mathbf{g}^{-1}(P) \cup \{\alpha\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) \wedge a \in \alpha_{\text{end}}(\alpha) \wedge \alpha \not\rightsquigarrow_i p).$$

Corollary 13 shows that each world which admits a regularity of the kind  $\forall i : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \not\rightsquigarrow_i p$  (for some  $\mathbf{g}, a, p$ ) admits an anonymous, conclusive claim that holds and is permanently verifiable under  $(\neg \text{POF}_C) \cap E \cap D^{\mathbf{g},a,p}$ .

*Example 3.* Let us illustrate this latest axiom and also the fact that a claim need not pinpoint every possible regularity to be useful. Suppose that the world contains two kinds of individuals. The jays find  $a$  decisive for  $\varphi$ . The kays however find  $b$  decisive for  $\varphi'$ . Assume some researcher becomes aware of a part of this regularity: she realizes that individuals who, upon exposure to  $(a, b)$ , answer  $\varphi$ , all satisfy  $a \hookrightarrow_i \varphi$ , but ignores that a similar regularity holds for the kays. She claims  $C = [\forall i : (a, b) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi]$ . This claim is anonymous and conclusive, and permanently verifiable under the axiom  $\neg \text{POF}_C \cap E \cap D^{\mathbf{g},a,\varphi}$ , with  $\mathbf{g} = \{(a, b), \varphi\}$ . As the axiom holds, this implies that the claim holds. The axiom  $D^{\mathbf{g},a,\varphi}$  requires to accept that whenever some individual satisfies  $(a, b) \rightsquigarrow_i \varphi$  but fails  $a \hookrightarrow_i \varphi$ , some path that completes  $(a, b)$  shows it, such as  $(a, b, c, a) \not\rightsquigarrow_i \varphi$ . On the contrary, the axiom fails if the only paths where  $a$  fails to be decisive for that individual are paths that do not complete  $(a, b)$ . This could be the case, for example, if that individual is definitely impressed by the first influence he is exposed to, and never considers any other influences afterwards.  $\triangle$

A sufficient (but not necessary) condition for this axiom to be satisfied is that repeating often enough the influences that lead the individual out of  $p$  lets her forget about the influences used in  $\mathbf{g}$ .

## 5. Conclusion

This study shows that, expectedly, general claims about regularity of behaviors of individuals facing exposure to influences can be seen to hold only under some prior knowledge, phrased here as axioms, that requires acceptance beyond empirical verifiability. Without delving into epistemological analysis, which I am not qualified to give, let us note that this is not specific to our endeavor and should come as no surprise: many claims in science require some non empirically verifiable assumptions for full applicability. For example, the apparently very low risk claim that the law of gravity applied

in my living room yesterday, even though I was not there to observe it, requires some regularity assumption; claiming on the contrary that yesterday in my living room was an exception to the law of gravity (but that it holds everywhere else) is compatible with all observations done so far to validate (or attempt to falsify) it. (This is a variant of the argument of Goodman [1983] about grass being green until some year and blue afterwards.)

This article has defined the concept of deliberated judgment and related notions, and pinpointed the precise axioms whose acceptance is required in order to be convinced of the veracity of a claim about deliberated judgments.

As example 3 illustrates, an axiom may be hard to accept, depending on the exposure protocol and what is considered plausible behavior. The framework presented here is general and simple, a good thing for an initial study of a concept: complexity requires justification. But this framework may lack richness for some exposure protocols. For example, it may be visible in the content of an influence (say, of an argument) that it supports  $\varphi$ , thus, if an individual is in favor of  $\varphi'$  after having been exposed to that influence, it may be considered a priori convincing that the individual would have been in favor of  $\varphi'$  also without seeing the influence. Our framework forbids such prior knowledge about the influences. Supplementary structure relaxing this could represent a middle-ground between our completely opaque approach of influences and the whole logical apparatus of formal argument theory which does not consider the argument's compatibilities as empirical, and would be worth studying.

The dependence of the deliberated judgment concept on the exposure protocol yields a qualification about the concept of deliberated judgments when one desires to use it normatively: it is only as reasonable to call the resulting judgments “deliberate” in a normative sense as the exposure protocol is normatively relevant to deliberation. A sufficiently restricted, or even purposefully chosen, set of influences may yield judgments that are only deliberate in the technical sense used here, but that pertain more to brain-washing than to reflective judgments. While it is indeed a long-term goal of this research program to yield a normatively relevant conception of preference (appropriate, for example, for recommendation), it is not a claim of this article that anything that corresponds to deliberated judgments as defined here qualify as being appropriate for any normative use: much depends on the chosen exposure protocol. The goal of this article is rather one of a conceptual and practical clarification. It proposes a sharp distinction between the empirical content and the normative content of the concept of deliberated judgments: the definition proposed here transfers the full normative weight on the choice of the exposure protocol; and leaves the rest of the disagreements to empirical settling. This is methodologically akin, *mutatis mutandis*, to the axiomatic study of justice such as done in social choice, only complementing mathematical deduction with empirical claims. In social choice, the choice of axioms bears the full normative weight; once axioms are chosen, it is a matter of logic which social choice rules (if any) correspond to that choice. In our case, once an exposure protocol is chosen, it is an empirical matter which deliberated judgments result. In supplement to a possibly helpful conceptual clarification, such a sharp separation may reveal practically useful for separation of concern as it permits to study deliberated judgments empirically under various exposure protocols independently

of possible disagreements about which norms are more appropriate for deliberation in which circumstances.

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## A. Proofs and supplementary discussion

### A.1. Generalities

This appendix relaxes the assumption of the main text that  $\#I \notin \mathbb{N}$  and  $\#\mathcal{A} \notin \mathbb{N}$ . It also relaxes the requirement that a claim be non-empty, thus, in this part, a claim is any subset of  $P^{I \times \mathcal{A}^{<\mathbb{N}}}$ . Each result indicates which assumptions on the cardinalities of these sets (or whether  $C \neq \emptyset$ ) are required.

We say that a claim  $C$  *holds* iff  $\rightsquigarrow \in C$ ;  $C$  *fails* otherwise.

The claim  $C$  is verifiable in the world  $\rightsquigarrow$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \uparrow \theta \subseteq C$ .  $C$  is possibly verifiable iff it is verifiable in some world; formally, iff  $\exists \theta \in \Theta \mid \uparrow \theta \subseteq C$ . Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ,  $C$  is verifiable at  $\rightsquigarrow$  under axiom  $D$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \emptyset \neq \uparrow \theta \cap D \subseteq C$ , and possibly verifiable under axiom  $D$  iff  $\exists \theta \in \Theta \mid \emptyset \neq \uparrow \theta \cap D \subseteq C$ . Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  and  $\rightsquigarrow \in C \cap D$ ,  $C$  is *permanently verifiable* (PV) under  $D$  at  $\rightsquigarrow$  iff  $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \uparrow \theta' \cap D \subseteq C$ .

Established verifiability (EV) mandates verifiability in all compatible worlds; formally,  $\forall \rightsquigarrow \in C : \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \sqsubseteq \rightsquigarrow' : \rightsquigarrow' \in C$ .

*Example 4.* The claim  $C$  being  $[\forall i : (a) \rightsquigarrow_i \varphi] \vee (a') \rightsquigarrow_i \varphi$  is possibly verifiable but fails established verifiability.  $\triangle$

A claim  $C$  is *possibly informative* iff it is informative in some world of  $C$ , namely, iff  $\exists \theta \in \Theta \cap C \subseteq \mid \pi(\uparrow \theta \cap C) \neq (P \cup \{\emptyset\})^I$ . A claim  $C$  is *informative* iff  $\pi(C) \neq (P \cup \{\emptyset\})^I$ . A claim  $C$  is *informative beyond counting* iff  $\exists i \in I \mid \pi_i(C) = \{\pi_i(\rightsquigarrow) \mid \rightsquigarrow \in C\} \neq P \cup \{\emptyset\}$ . A claim  $C$  is a *posteriori sharp* iff  $\exists \theta \in \Theta, i \in I \mid \pi_i(\uparrow \theta \cap C) = \#\{\pi_i(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\} = 1$ . A claim  $C$  is a *posteriori truly sharp* iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), i \in I \mid \pi_i(\uparrow \theta \cap C) = \#\{\pi_i(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\} = 1$ .

*Remark 3.* Informativeness permits mere counting: the claim that exactly two persons deliberately believe  $\varphi$  is informative. Also, the claim  $C = \{[\exists i \mid (a) \not\rightsquigarrow_i \varphi] \wedge [\forall i : (a) \not\rightsquigarrow_i \varphi \Rightarrow \varphi' \in \pi_i(\rightsquigarrow)]\}$  is informative. Informativeness may be a desirable property (rather than informativeness beyond counting) as one may, for example, want to know whether a given proposition is believed by a majority of individuals. Note that informativeness beyond counting implies informativeness.  $\triangle$

Note that  $C^\subseteq$  is downwards closed, meaning that  $\forall c \in C^\subseteq : c' \subseteq c \Rightarrow c' \in C^\subseteq$  (equivalently,  $\{c' \subseteq c\} \subseteq C^\subseteq$ ). Both  $\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I$  and  $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$  are downwards closed. The set  $\Theta \cap C^\subseteq$  is downwards closed, as both  $\Theta$  and  $C^\subseteq$  are.

*Example 5.* Given  $a, i$  and  $\varphi$ , the claim  $[a \rightsquigarrow_i \varphi]$  is not anonymous iff  $\#I \geq 2$ . Indeed, pick any  $j \neq i$  and define  $\sigma(i) = j, \sigma(j) = i$  and  $\forall k \in I \setminus \{i, j\} : \sigma(k) = k$ , then  $C = \{\rightsquigarrow \mid \rightsquigarrow(i, a) = \varphi\}$ , which contains in particular  $\rightsquigarrow$  defined as  $\forall \alpha : [\forall k \in I \setminus \{j\}, \rightsquigarrow(k, \alpha) = \varphi \wedge \rightsquigarrow(j, \alpha) = \varphi']$ ; while  $C \circ \sigma = \{\rightsquigarrow \mid \rightsquigarrow(j, a) = \varphi\}$  which does not contain  $\rightsquigarrow$ .  $\triangle$

## A.2. Verifiability

**Theorem 2.** *Given an axiom  $D$  such that  $\rightsquigarrow \in D$ , if a claim  $C$  is verified under  $D$ , then  $C$  holds.*

*Proof.* Consider any  $\theta$  that satisfies verifiedness under  $D$ . We have  $\theta \subseteq \rightsquigarrow$ . By hypothesis,  $\rightsquigarrow \in D$ . Therefore, by verifiedness,  $\rightsquigarrow \in C$ .  $\square$

Using the trivial axiom  $D = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , we obtain this corollary.

**Corollary 1.** *If a claim is verified, it holds.*

**Theorem 3.** *For any  $\theta \in \Theta$  and claims  $C, D$ , these properties are equivalent:*

1.  $\forall \theta \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$ ,
2.  $\uparrow \theta \cap D \subseteq C$ ,
3.  $\theta \notin (D \setminus C)^\subseteq$ ,
4. *there exists a finite disjoint partition  $\{S_k\}$  of  $D \setminus C$  such that  $\forall k, \exists t \in \theta \mid \{t\} \notin S_k^\subseteq$  (equivalently,  $t \notin \bigcup S_k$ ).*

*Proof.* Properties 1 and 2 assert that  $\forall \theta \subseteq \rightsquigarrow : \rightsquigarrow \in D \Rightarrow \rightsquigarrow \in C$ , equivalently,  $\nexists \theta \subseteq \rightsquigarrow \mid \rightsquigarrow \in D \wedge \rightsquigarrow \notin C$ , which is property 3.

To establish that property 3 implies property 4, consider any claim  $E$  and  $\theta \notin E^\subseteq$ . Given any  $t \in \theta$ , define  $S_t = \{\rightsquigarrow \mid t \notin \rightsquigarrow\} \cap E$ . Define  $S = \bigcup_{t \in \theta} S_t$ .

Consider any  $\rightsquigarrow \in E$ . As  $\theta \not\subseteq \rightsquigarrow$ ,  $\exists t \in \theta \mid t \notin \rightsquigarrow$ , thus,  $\rightsquigarrow \in S$ . It follows that  $S = E$ .

As  $\theta$  is finite, we can finitely enumerate the elements of  $\theta = \{t_k, 1 \leq k \leq \#\theta\}$ , and define  $S_1 = S_{t_1}$  and for any  $2 \leq k \leq \#\theta$ ,  $S_k = S_{t_k} \setminus \bigcup_{1 \leq k' < k} S_{t_{k'}}$ . We obtain  $E = S = \bigcup_{1 \leq k \leq \#\theta} S_k$ , thus it is a disjoint partition of  $E$ , and as  $\{t_k\} \notin S_{t_k}^\subseteq$  by construction of  $S_{t_k}$ , also  $\{t_k\} \notin S_k^\subseteq$ .

Finally, negating property 3, if  $\theta$  is completable to some set covered by  $\{S_k\}$ , then  $\theta$  is completable to some set  $S_k$ , which contradicts property 4.  $\square$

**Lemma 1.** *If  $\#\mathcal{A} \notin \mathbb{N}$ , given any general observation  $O$  and  $f : I \rightarrow (P \cup \{\emptyset\})$ ,  $\exists \rightsquigarrow \in P^{I \times \mathcal{A} < \mathbb{N}} \mid O \subseteq \rightsquigarrow \wedge \pi(\rightsquigarrow) = f$ .*

*Proof.* For each  $i \mid f(i) \in P$ , define  $a_i \notin (O(P))(i)(\mathbb{N})$  as an influence not used for  $i$ ; for each  $i \mid f(i) = \emptyset$ , for each  $p \in P$ , define  $k_i^p \notin (O(P))(i)$  as a path not used for  $i$ , with all  $k_i^p$  different from each other ( $\forall p, p' \in P, k_i^p \neq k_i^{p'}$ ).

Define  $f' = O \cup_i \mid f(i) \in P, \alpha \mid a_i \in \alpha_{end} \{(i, \alpha, f(i))\} \cup_i \mid f(i) = \emptyset, p \in P, a \in \mathcal{A} \{(i, (k_i^p, a), p)\}$ . Define  $\rightsquigarrow$  as any completion of  $f'$  to a total function in  $P^{I \times \mathcal{A} < \mathbb{N}}$ .  $\square$

**Theorem 4.** *If  $C$  is such that  $\pi(C) \neq (P \cup \{\emptyset\})^I$  and is possibly verifiable under  $D$ , then*

- $\exists \theta \in \Theta, f \in (P \cup \{\emptyset\})^I \mid \forall \rightsquigarrow \in D : [\theta \subseteq \rightsquigarrow \Rightarrow \pi(\rightsquigarrow) \neq f]$ ;
- *if  $\#\mathcal{A} \notin \mathbb{N}$ ,  $D$  is not trivial.*

*Proof.* By theorem 3,  $\uparrow \theta \cap D \subseteq C$ , and for some  $f, \rightsquigarrow \in C \Rightarrow \pi(\rightsquigarrow) \neq f$ .

Also,  $D \neq P^{I \times \mathcal{A} < \mathbb{N}}$ , as  $\exists \theta \subseteq \rightsquigarrow \mid \pi(\rightsquigarrow) = f$  (by lemma 1), thus  $\rightsquigarrow \notin D$ .  $\square$

**Corollary 2.** *If  $\#\mathcal{A} \notin \mathbb{N}$ , any informative claim is not possibly verifiable.*

**Theorem 5.** *If  $\#\mathcal{A} \notin \mathbb{N}$ , any claim verifiable in  $\rightsquigarrow$  is informative iff it is informative in  $\rightsquigarrow$ .*

*Proof.* Let  $\theta_v \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  witness that  $C$  is verifiable in  $\rightsquigarrow$ , so that every world extending  $\theta_v$  lies in  $C$ ; in particular  $\rightsquigarrow \in C$ , hence informativeness in  $\rightsquigarrow$  is defined.

If  $C$  is informative, then  $\emptyset \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  witnesses that it is informative in  $\rightsquigarrow$ , since  $\uparrow \emptyset \cap C = C$  and thus  $\pi(\uparrow \emptyset \cap C) = \pi(C) \neq (P \cup \{\emptyset\})^I$ .

Conversely, assume  $C$  is not informative, that is,  $\pi(C) = (P \cup \{\emptyset\})^I$ ; we show it is not informative in  $\rightsquigarrow$ . Let  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ . As  $\theta \subseteq \rightsquigarrow$  and  $\theta_v \subseteq \rightsquigarrow$ , the union  $\theta \cup \theta_v$  is a right unique finite subset of  $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$ , hence a general observation. Given any  $f \in (P \cup \{\emptyset\})^I$ , lemma 1 yields  $\rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$  with  $\theta \cup \theta_v \subseteq \rightsquigarrow'$  and  $\pi(\rightsquigarrow') = f$ . Then  $\theta_v \subseteq \rightsquigarrow'$  gives  $\rightsquigarrow' \in C$ , and  $\theta \subseteq \rightsquigarrow'$  gives  $\rightsquigarrow' \in \uparrow \theta$ , so  $f \in \pi(\uparrow \theta \cap C)$ . As  $f$  is arbitrary,  $\pi(\uparrow \theta \cap C) = (P \cup \{\emptyset\})^I$ ; as  $\theta$  is arbitrary,  $C$  is not informative in  $\rightsquigarrow$ .  $\square$

*Remark 4.* The hypothesis  $\#\mathcal{A} \notin \mathbb{N}$  cannot be dropped: without it, a claim may be verifiable and informative in  $\rightsquigarrow$  yet not informative (the left to right direction however, informative implies informative in a given world, uses no such hypothesis). Let  $\mathcal{A} = \{a\}$  and  $I = \{i_0\}$ , so that every nonempty path ends in  $\{a\}$  and  $\pi(\rightsquigarrow)(i_0) = \{p\}$  if  $\rightsquigarrow$  responds  $p$  to every nonempty path and  $\emptyset$  otherwise. Write  $\rightsquigarrow_p$  for the world responding  $p$  everywhere, let  $\theta_v = \{(i_0, (a), \varphi')\}$ , and let  $C = \uparrow \theta_v \cup \{\rightsquigarrow_\varphi, \rightsquigarrow_0\}$ . Then  $C$  is verifiable in  $\rightsquigarrow_{\varphi'}$ , witnessed by  $\theta_v$ . It is also informative in  $\rightsquigarrow_{\varphi'}$ : as  $\rightsquigarrow_\varphi, \rightsquigarrow_0 \notin \uparrow \theta_v$ , we have  $\uparrow \theta_v \cap C = \uparrow \theta_v$ , and forcing  $(a) \rightsquigarrow_{i_0} \varphi'$  leaves only  $\pi(\rightsquigarrow)(i_0) \in \{\{\varphi'\}, \emptyset\}$  (reaching  $\{\varphi\}$  or  $\{0\}$  would require every nonempty path to answer  $\varphi$ , resp. 0), so  $\pi(\uparrow \theta_v \cap C) = \{\{\varphi'\}, \emptyset\} \neq (P \cup \{\emptyset\})^I$ . Yet  $C$  is not informative, since  $\rightsquigarrow_\varphi$  and  $\rightsquigarrow_0$  supply the missing profiles:  $\pi(C) = \{\{\varphi'\}, \emptyset, \{\varphi\}, \{0\}\} = (P \cup \{\emptyset\})^I$ .  $\triangle$

### A.3. Falsifiability

The following partitioning of claims with respect to their falsifiability status and examples might help better understand this central notion.

Any claim  $C$  satisfies exactly one of these:

**falsifiable** iff  $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$  and  $\exists \rightsquigarrow \notin C$ ;

**never falsifiable** iff  $\forall \rightsquigarrow \notin C : \neg \text{OF}_C(\rightsquigarrow)$ , equivalently, iff  $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$ , equivalently, iff  $\Theta \subseteq C^\subseteq$ ;

**with mixed falsifiability** iff  $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$  and  $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$ .

Indeed, first implies  $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$  which negates second, and the negation of the conjunction of the first two is  $(\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow) \vee \nexists \rightsquigarrow \notin C) \wedge (\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow))$ , equivalent to the third.

We want to study the following dichotomy. Any claim  $C$  satisfies exactly one of these:

**falsifiable or trivial** iff  $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$ , equivalently, iff  $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$ , still equivalently, iff  $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow)$ ;

**not falsifiable and not trivial** iff  $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$ , that is, whenever the implication  $F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$  fails.

Distinguishing two cases among the latter situation leads to the following partition. Any claim  $C$  satisfies exactly one of these:

**falsifiable or trivial** iff  $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$ , equivalently, iff  $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$ , still equivalently, iff  $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow)$ ;

**never falsifiable and not trivial** iff  $\forall \rightsquigarrow \notin C : \neg \text{OF}_C(\rightsquigarrow)$  and  $\exists \rightsquigarrow \notin C$ ;

**with mixed falsifiability** iff  $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$  and  $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$ .

Indeed, second implies  $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$  which negates first, and the negation of the conjunction of the first two is  $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow) \wedge (\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow) \vee \nexists \rightsquigarrow \notin C)$ , equivalent to the third.

Here are typical examples.

**Falsifiable**  $(a) \rightsquigarrow_i \varphi$  (satisfies established verifiability);  $\forall i : (a) \rightsquigarrow_i \varphi$  (fails possible verifiability);  $\forall i : a \hookrightarrow_i \varphi$  (fails possible verifiability) (example 7).

**Never falsifiable**  $[\exists i \mid (a) \rightsquigarrow_i \varphi]$  which satisfies established verifiability;  $[\forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]]$  (example 6) which fails possible verifiability;  $[(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$  which satisfies established verifiability;  $[\forall i : (a) \rightsquigarrow_i \varphi] \vee [\forall i : (a, a) \rightsquigarrow_i \varphi] \vee \dots$  (example 8) which fails possible verifiability.

**With mixed falsifiability**  $[\exists i \mid (a) \rightsquigarrow_i \varphi] \wedge (a') \rightsquigarrow_i \varphi$  (proof:  $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow) \Leftrightarrow (a') \not\rightsquigarrow_i \varphi$ ) (satisfies established verifiability);  $[\exists i \mid (a) \rightsquigarrow_i \varphi] \wedge [\forall i : (a') \rightsquigarrow_i \varphi]$  (proof:  $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow) \Leftrightarrow [\exists i \mid (a') \not\rightsquigarrow_i \varphi]$ ) (fails possible verifiability); “Exactly two of the three paths in  $\{\emptyset, (a), (a')\}$  lead  $i$  to  $\varphi$  and the remaining one leads  $i$  to  $\varphi'$ ” (example 10) which fails possible verifiability.

We also say that  $C$  is *rejectable* iff  $\text{OF}_C(\rightsquigarrow)$ ; and  $C$  is *persistently falsifiable* iff  $C \neq P^{I \times \mathcal{A}^{<\mathbb{N}}} \wedge [\forall \rightsquigarrow : F_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)]$ .

*Example 6* (Disjunct with incompatible paths, not rejectable). Given  $a \neq a' \in \mathcal{A}$ , the claim  $\forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$  is not rejectable. In fact,  $\Theta \subseteq C^\subseteq$ . Consider any  $\theta \in \Theta$  and let us prove that  $\theta \in \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$ . By definition of  $\Theta$ ,  $\theta(P)$  may not contain both  $((a), i)$  and  $((a'), i)$ , thus letting  $t$  denote one of these pairs not in  $\theta(P)$  and defining  $r = (t, \varphi)$ , pick any  $\rightsquigarrow'$  such that  $\theta \subseteq \rightsquigarrow' \wedge t \in \rightsquigarrow'$ . We have  $\theta \in \mathcal{P}(\rightsquigarrow')$  and  $\rightsquigarrow' \in C$ .  $\triangle$

Example 6 shows a claim that is not rejectable, irrespective of the exposure protocol. In particular, the example applies when  $C$  does not hold (that is, when  $\rightsquigarrow \notin C$ ). We conclude that a claim may fail to hold but still be not rejectable. It follows that non-rejection cannot be a sufficient condition for validating a claim.

*Remark 5.* Examples 8 and 10 present situations where, similarly, the claim may fail to hold but still be not rejectable. In example 8, non rejection is due to finiteness of observations. In example 10, rejection depends on  $\rightsquigarrow$ .  $\triangle$

*Example 7.* Consider any  $a$  and  $\varphi$ . The claim  $\forall i : a \hookrightarrow_i \varphi$  is falsifiable: given any  $\rightsquigarrow \notin C$ , consider any  $i \mid a \not\rightsquigarrow_i \varphi$  and any  $\alpha \mid \alpha \not\rightsquigarrow_i \varphi \wedge a \in \alpha_{\text{end}}$ , and define  $\theta = \{(i, \alpha, \rightsquigarrow(i, \alpha) \neq \varphi)\}$ .  $\triangle$

The following example presents a situation where non rejection is due to finiteness of observations.

*Example 8* (Big disjunct, non rejectable). Consider  $a \in \mathcal{A}$ . The claim  $[\forall i : (a) \rightsquigarrow_i \varphi] \vee [\forall i : (a, a) \rightsquigarrow_i \varphi] \vee \dots$ , thus,  $C = \{\rightsquigarrow \mid \exists n \in \mathbb{N}^* \mid \forall i : (a)^n \rightsquigarrow_i \varphi\}$ , is non rejectable:  $\Theta \subseteq C^\subseteq$ .

Given any  $\theta \in \Theta$ , define  $l = \max_{\alpha \mid \exists i \in I, p \in P \mid (i, \alpha, p) \in \theta} \# \alpha + 1$  if  $\exists (i, \alpha, p) \in \theta$ , and  $l = 1$  otherwise. Pick any  $\rightsquigarrow$  such that  $\theta \subseteq \rightsquigarrow \wedge (a)^l \rightsquigarrow_i \varphi$ . We have  $\theta \in \mathcal{P}(\rightsquigarrow)$  and  $\rightsquigarrow \in C$ , thus,  $\theta \in C^\subseteq$ .  $\triangle$

*Example 9* (Big disjunct unfals iff infinite). Given any  $a \in \mathcal{A}$  and an infinite  $I$ , the claim  $\exists i \mid a \hookrightarrow_i \varphi$  is strongly unfalsifiable: any  $\theta \in \Theta$  only has finitely many  $i$  in its triples, thus, some  $i$  is not reached by  $\theta$ , thus  $\theta \in C^\subseteq$ . If  $I$  is finite, then the claim is falsifiable: if it is wrong, observe for each  $i$  some adequate triple  $(i, \alpha_i, p \neq \varphi)$ .  $\triangle$

**Theorem 6.**  $\forall C, \rightsquigarrow \in C : \Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$ .

*Proof.* This holds by definition of  $C^\subseteq = \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$ .  $\square$

*Example 10* (Three functions, may be rejectable). Consider  $i \in I$ ,  $a \neq a' \in \mathcal{A}$  and the three paths  $B = \{\emptyset, (a), (a')\}$ . The claim “Exactly two of the three paths in  $B$  lead  $i$  to  $\varphi$  and the remaining one leads  $i$  to  $\varphi'$ ” is not rejectable iff  $\rightsquigarrow_i$  associates, in order,  $(\emptyset, (a), (a'))$  to either  $(\varphi', \varphi, \varphi)$ ,  $(\varphi, \varphi', \varphi)$ ,  $(\varphi, \varphi, \varphi')$ ,  $(\varphi, \varphi, \varphi)$  or  $(\varphi, \varphi', \varphi')$ .

Indeed, if  $\rightsquigarrow_i$  satisfies this, then either it holds and by theorem 6 it is not rejectable, or it associates this triplet to  $(\varphi, \varphi, \varphi)$  or  $(\varphi, \varphi', \varphi')$  in which case, as  $(a)$  and  $(a')$  cannot be together in  $\theta$ , any  $\theta \in \Theta$  with  $\theta \subseteq \rightsquigarrow$  satisfy  $\theta \in C^\subseteq$ .

Conversely, if  $C$  is not rejectable, then for any  $\theta_1$  containing  $\emptyset$  and  $(a)$  and  $\theta_2$  containing  $\emptyset$  and  $(a')$ , with  $\theta_1, \theta_2 \subseteq \rightsquigarrow$ , we have  $\theta_1, \theta_2 \in C^\subseteq$ , which requires that  $\rightsquigarrow_i$  associates  $(\emptyset, (a))$  to  $(\varphi', \varphi)$ ,  $(\varphi, \varphi')$  or  $(\varphi, \varphi)$  and  $(\emptyset, (a'))$  to  $(\varphi', \varphi)$ ,  $(\varphi, \varphi)$  or  $(\varphi, \varphi')$ .  $\triangle$

**Theorem 7.** Given any claims  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ,  $[\rightsquigarrow \in D \wedge \text{OF}_{D \setminus C}(\rightsquigarrow)] \Leftrightarrow [\rightsquigarrow \in C \cap D \wedge \text{OF}_{D \setminus C}(\rightsquigarrow)]$ .

*Proof.* Given any  $\rightsquigarrow$ ,  $\text{OF}_{D \setminus C}(\rightsquigarrow) \Rightarrow \rightsquigarrow \notin D \setminus C$ ; equivalently,  $\text{OF}_{D \setminus C}(\rightsquigarrow) \Rightarrow \rightsquigarrow \in C \cup \overline{D}$ .  $\square$

**Corollary 3.** Given any claims  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , established verifiability under axiom  $D$ , that is,  $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$ , is equivalent to  $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin (D \setminus C)^\subseteq$ .

**Corollary 4.**  $C$  satisfies established verifiability under axiom  $D$  iff  $\forall \rightsquigarrow \in C \cap D : \text{OF}_{D \setminus C}(\rightsquigarrow)$ .

Theorem 10 shows that falsifiability indeed fills the mentioned gap: falsifiability guarantees that a non trivial claim holds if, and only if, it is not rejectable.

*Remark 6.* Making a claim more precise (meaning going from a claim  $C$  to  $C' \subsetneq C$ ) can make it falsifiable, or conversely, can make it unfalsifiable. Given  $\alpha, i$ , consider the claim  $C_2 = \alpha \rightsquigarrow_i \varphi$ . It is falsifiable. Now consider the claim  $C = C_1 \cap C_2$  with  $C_1$  being the claim “every individual chooses  $\varphi$  after seeing solely either  $a$  or  $a'$ ”. It is more precise but unfalsifiable. But we can make it falsifiable again by making it even more precise (picking any single relation).  $\triangle$

**Theorem 8.** *Given any claim  $C$ , the axiom  $\neg \text{OF}_C(\rightsquigarrow)$  proves  $C$  iff  $C$  is falsifiable or trivial.*

*Proof.*  $D$  proves  $C$  means, by definition,  $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow) \Rightarrow \neg F_C(\rightsquigarrow)$ .  $\square$

Provability by  $\neg \text{OF}_C(\rightsquigarrow)$  (equivalently, falsifiability or triviality) is strictly stronger than established verifiability under  $\neg \text{OF}_C(\rightsquigarrow)$ ; and established verifiability is strictly stronger than established verifiability under  $\neg \text{OF}_C(\rightsquigarrow)$ .

*Example 11.* The claim  $C_1 = \{\exists i \mid (a) \rightsquigarrow_i \varphi\}$ , with infinitely many individuals, satisfies established verifiability (consider  $\theta = \{(a) \rightsquigarrow_i \varphi\}$ ), hence, established verifiability under  $\neg \text{OF}_C(\rightsquigarrow)$ , but is not provable by  $\neg \text{OF}_C(\rightsquigarrow)$ . Similarly for  $C_2 = \{(a) \rightsquigarrow_i \varphi \vee (b) \rightsquigarrow_i \varphi\}$  and  $\theta = \{(a) \rightsquigarrow_i \varphi\}$ .

Also, consider  $C_3$  being  $\forall j : (c) \rightsquigarrow_j \varphi$ ,  $C_4$  being  $(d) \rightsquigarrow_j \varphi \wedge (e) \rightsquigarrow_j \varphi$  and  $C$  being  $C_1$  or  $C_2$  together with  $C_3$  or  $C_4$  (assuming infinitely many individuals if using  $C_1$  or  $C_3$ ). Then,  $C$  fails established verifiability (due to  $C_3$  or  $C_4$ ) and  $\neg \text{OF}_C(\rightsquigarrow)$  does not prove  $C$  (due to  $C_1$  or  $C_2$ ), but  $C$  satisfies established verifiability under axiom  $\neg \text{OF}_C(\rightsquigarrow)$  (with verification  $(a) \rightsquigarrow_i \varphi$ ).  $\triangle$

This suggests conjecturing that established verifiability under  $D'$  is always strictly stronger than established verifiability under  $D$ , for any axioms  $D \subseteq D'$ .

Note that if a claim is not provable by  $\neg \text{OF}_C(\rightsquigarrow)$  (equivalently, some world is out of  $C$  and NOT OF), then some world is out of  $C$  and NOT POF.

It is probably useful to remember that  $\neg \text{OF}_C(\rightsquigarrow)$  states that negating  $C$ , in that world, requires incompatible, or infinitely many, triples: the triples completable only out of  $C$  are all unobservable. Indeed,  $\Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$  is equivalent to  $\forall T \in \mathcal{P}(\rightsquigarrow), T \notin C^\subseteq \Rightarrow T \notin \Theta$ . When a claim is never falsifiable, thus  $\forall \rightsquigarrow, \neg \text{OF}_C(\rightsquigarrow)$ , we obtain  $\Theta \subseteq C^\subseteq$ .

If  $D \subseteq D'$ , then established verifiability under axiom  $D'$  implies established verifiability under axiom  $D$ .

Corollary 10 shows that impossibilities propagate to stricter claims. Together with theorem 4, it shows that the axiom required must be at least as strong as claiming that if some observable holds, then some configuration of DJs is excluded.

*Remark 7.* Suffices that the axiom states something like: if almost all individuals deliberately believe  $\varphi$ , then everybody deliberately believe  $\varphi$ . This innocuous enough axiom suffices because we request only informativeness, and indeed it only permits to restrict possibilities, not to determine anybody's DJ.  $\triangle$

## A.4. Verifiability under an axiom

**Theorem 9.** *Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  and  $\rightsquigarrow \in C \cap D$ :  $C$  is PV under  $D$  at  $\rightsquigarrow$  iff  $\text{POF}_{D \setminus C}(\rightsquigarrow)$ .*

*Proof.* By theorem 3 (items 1 and 3), for any  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ :  $\theta'$  proves  $C$  under  $D$  (that is,  $\forall \theta' \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$ ) iff  $\theta' \notin (D \setminus C)^\subseteq$ . Substituting into the definition of PV at  $\rightsquigarrow$  gives  $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \notin (D \setminus C)^\subseteq$ , which is exactly  $\text{POF}_{D \setminus C}(\rightsquigarrow)$ .  $\square$

**Corollary 5.**  *$C$  satisfies PV under axiom  $D$  iff  $\forall \rightsquigarrow \in C \cap D : \text{POF}_{D \setminus C}(\rightsquigarrow)$ .*

**Corollary 6.** *Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ : PV under  $D$  implies established verifiability under  $D$ .*

*Proof.*  $\text{POF}_S(\rightsquigarrow) \Rightarrow \text{OF}_S(\rightsquigarrow)$  for any  $S, \rightsquigarrow$  (taking  $\theta = \emptyset$  in the definition of  $\text{POF}_S$  already gives the witness required by  $\text{OF}_S$ ), so corollary 5 together with corollary 4 gives the first implication.  $\square$

**Corollary 7.** *Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ : provability of  $C$  by  $D$  implies PV under  $D$ .*

*Proof.* If  $D \subseteq C$ , then  $D \setminus C = \emptyset$ , so  $(D \setminus C)^\subseteq = \emptyset$  and  $\text{POF}_\emptyset(\rightsquigarrow)$  holds trivially (any  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ , in particular  $\theta' = \emptyset$ , already satisfies  $\theta' \notin \emptyset$ ), giving PV under  $D$  at every  $\rightsquigarrow \in C \cap D$  by corollary 5.  $\square$

## A.5. Observational exclusion and completion

$S \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  is observationally excludable iff  $\exists \theta \in \Theta \mid \theta \notin S^\subseteq$ , equivalently, iff  $\Theta \not\subseteq S^\subseteq$ , equivalently, iff  $\exists \rightsquigarrow \notin S \mid \text{OF}_S(\rightsquigarrow)$ , equivalently, iff  $\exists \rightsquigarrow \mid \text{OF}_S(\rightsquigarrow)$ , equivalently, iff  $S$  is NOT never falsifiable.

$D$  is observationally completable to  $C$  iff  $\exists \theta \in (\Theta \cap D^\subseteq) \mid \theta \notin (D \setminus C)^\subseteq$ . Corollary 9 shows the equivalence with possible verifiability under axiom  $D$ .

$D$  is necessarily observationally completable to  $C$  iff  $\forall \rightsquigarrow \in D, \text{OF}_{D \setminus C}(\rightsquigarrow)$ . Remark 13 shows the lack of equivalence with established verifiability under axiom  $D$ .

Converting a claim to disjunctive normal form is useful to characterize its compatible observations, as theorem 20 indicates.

*Example 12.* Consider six different influences  $a_1, a'_1, a_2, a'_2, a_3, a'_3 \in \mathcal{A}$  and the following claim  $E$ : if any two influences  $a_k, a_{k'}$  fail to convince  $i$  or  $a'_k, a'_{k'}$  fail to convince  $j$ , then the third influence  $a_{k''}$  convinces  $i$  and  $a'_{k''}$  convinces  $j$ . The claim is not observationally excludable. Indeed, we see that  $E = \{\rightsquigarrow \mid [(a_1) \rightsquigarrow_i \varphi \wedge (a'_1) \rightsquigarrow_j \varphi] \vee [(a_2) \rightsquigarrow_i \varphi \wedge (a'_2) \rightsquigarrow_j \varphi] \vee [(a_3) \rightsquigarrow_i \varphi \wedge (a'_3) \rightsquigarrow_j \varphi]\}$ , and  $\forall 1 \leq k \leq 3$ , define  $S_k = \{\rightsquigarrow \mid [(a_k) \rightsquigarrow_i \varphi \wedge (a'_k) \rightsquigarrow_j \varphi]\}$ , define  $\mathcal{S} = \{S_1, S_2, S_3\}$ , consider any  $f \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid f \notin E^\subseteq$ , we have  $\forall 1 \leq k \leq 3, \exists t_k \in f \mid \{t_k\} \notin S_k^\subseteq$ , thus  $f \notin \Theta$  (wlog, pick  $t_1 = ((a_1), i, p_1 \neq \varphi)$ , then for observability we need to pick  $t_2 = ((a'_2), j, \cdot)$  and there is no choice left for  $t_3$ ).  $\triangle$

$C$ -scrutiny is the axiom  $D_C = \{\rightsquigarrow \mid \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \in C^\subseteq\}$ .

*Remark 8.* Reusing, from remark 2, the example  $C_2 = [\forall i : (a') \rightsquigarrow_i \varphi \wedge (a) \rightsquigarrow_j \varphi]$  for a given  $j$  and considering again a world satisfying  $(a) \not\rightsquigarrow_j \varphi$  and  $\forall i : (a') \rightsquigarrow_i \varphi$ , we see that formalizing  $\neg \text{POF}_C$ , independently of  $\rightsquigarrow$ , as  $\exists \theta \in \Theta \mid \forall \theta' \sqsubseteq \theta' \in \Theta : \theta' \in C^\subseteq$ , is inadequate, as this prevents the prefix  $\theta = \{(a') \rightsquigarrow_j \varphi\}$  from bringing not POF. The condition  $\exists \theta \in \Theta \mid \forall \theta' \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \in C^\subseteq$  is also not suitable, as it is always satisfied: pick any  $\theta \in \Theta$  such that  $\theta \notin \mathcal{P}(\rightsquigarrow)$ .

The following alternative axiom, which might be considered for its apparent proximity with the chosen one, is also inadequate:  $\forall \rightsquigarrow, \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), [\exists \theta' \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta' \notin C^\subseteq] \Rightarrow [\theta \in C^\subseteq]$ . Consider again  $C_1$  and  $\rightsquigarrow$  such that  $(a) \rightsquigarrow_i \varphi \wedge (a') \not\rightsquigarrow_i \varphi$ . The chosen axiom ( $\text{POF}_{C_1}(\rightsquigarrow) \Rightarrow \neg \text{OF}_C(\rightsquigarrow)$ ) holds, as intuition commands, as the antecedent fails for the reason given above. The alternative axiom however fails as  $\theta = \{(a') \rightsquigarrow_i \varphi\}$  is completable only out of  $C$ .  $\triangle$

*Remark 9.* Verifiability and observable falsity have a dual character as  $C$  satisfies established verifiability under axiom  $D$  iff  $\forall \rightsquigarrow \in C \cap D, \text{OF}_{D \setminus C}(\rightsquigarrow)$  (corollary 4).  $\triangle$

**Theorem 10.** *For any falsifiable or trivial claim  $C$ :  $\rightsquigarrow \in C$  iff  $\neg \text{OF}_C(\rightsquigarrow)$ .*

This (to be proven) captures the claim “there is no decisive influence for  $i$ ”.

**Theorem 11.** *Given any  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ :  $\pi(\rightsquigarrow)(i) = \{\emptyset\} \Leftrightarrow \forall a \in \mathcal{A}, \exists p_1 \neq p_2 \in P, \alpha^1, \alpha^2 \in \mathcal{A}^{<\mathbb{N}} \mid a \in \alpha_{end}^1 \cap \alpha_{end}^2 \wedge \alpha^1 \rightsquigarrow_i p_1 \wedge \alpha^2 \rightsquigarrow_i p_2$ .*

*Example 13* (No decisive influence). The claim  $C = \{\rightsquigarrow \mid \pi(\rightsquigarrow)(I) = \{\emptyset\}\}$  is possibly verifiable under axiom  $D$  iff  $D$  is observationally completable to  $\{\rightsquigarrow \mid \pi(\rightsquigarrow)(I) = \{\emptyset\}\}$ .  $\triangle$

Theorem 11 and example 13 illustrate (what?) about no decisive influences.

**Theorem 12.**  *$D \setminus C$  is falsifiable or trivial iff ( $C$  satisfies established verifiability under axiom  $D$  AND  $\forall \rightsquigarrow \notin D : \text{OF}_{D \setminus C}(\rightsquigarrow)$ ).*

*Proof.*  $D \setminus C$  is falsifiable or trivial iff  $\forall \rightsquigarrow \notin (D \setminus C) : \text{OF}_{D \setminus C}(\rightsquigarrow)$ , equivalently (as  $\rightsquigarrow \notin D \setminus C \Leftrightarrow \rightsquigarrow \notin D \vee \rightsquigarrow \in C$ ), iff  $[\forall \rightsquigarrow \notin D : \text{OF}_{D \setminus C}(\rightsquigarrow)] \wedge [\forall \rightsquigarrow \in C : \text{OF}_{D \setminus C}(\rightsquigarrow)]$ . As  $C \setminus D \subseteq \overline{D}$ , the second conjunct is already implied on  $C \setminus D$  by the first, so it reduces to  $\forall \rightsquigarrow \in C \cap D : \text{OF}_{D \setminus C}(\rightsquigarrow)$ , that is, by corollary 4, to established verifiability under axiom  $D$ .  $\square$

**Corollary 8.**  *$P^{I \times \mathcal{A}^{<\mathbb{N}}} \setminus C$  is falsifiable or trivial iff  $C$  satisfies established verifiability.*

**Theorem 13.** *For any  $\theta \in \Theta$ , the following statements are equivalent.*

- $\theta \notin C^\subseteq$ ,
- $\forall \theta' \sqsubseteq \rightsquigarrow, \rightsquigarrow \notin C$ ,
- $\forall \theta' \sqsubseteq \rightsquigarrow : \text{OF}_C(\rightsquigarrow)$ ,
- $\theta \notin (\neg \text{OF}_C(\rightsquigarrow))^\subseteq$ .

*Proof.* 1 and 2 are equivalent by definition. From 1 to 3, that's because for any such  $\rightsquigarrow$ , we have indeed  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq$ .  $\square$

**Theorem 14.** *Given any claims  $C \subseteq C'$  and any  $\rightsquigarrow: \text{OF}_{C'}(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$ .*

*Proof.* By definition,  $\neg \text{OF}_C(\rightsquigarrow)$  means  $\Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$ , thus the contrapositive is proven by observing that  $C^\subseteq \subseteq C'^\subseteq$ .  $\square$

**Theorem 15.** *For any claim  $C$ , with  $D$  being  $\neg \text{OF}_C(\rightsquigarrow)$ , given any  $\rightsquigarrow: \text{OF}_C(\rightsquigarrow) \Leftrightarrow \text{OF}_D(\rightsquigarrow)$ ; and  $\text{OF}_D(\rightsquigarrow) \Rightarrow \text{OF}_{D \setminus C}(\rightsquigarrow)$  (using theorem 14).*

*Proof.*  $\text{OF}_C(\rightsquigarrow) \Leftrightarrow \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq \Leftrightarrow \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin (\neg \text{OF}_C(\rightsquigarrow))^\subseteq \Leftrightarrow \text{OF}_D(\rightsquigarrow) \Rightarrow \text{OF}_{D \setminus C}(\rightsquigarrow)$ .  $\square$

**Theorem 16.** *Considering a claim  $C$  and axiom  $D$  being  $\neg \text{OF}_C(\rightsquigarrow)$ ,  $D \setminus C$  is falsifiable or trivial iff  $C$  satisfies established verifiability under axiom  $\neg \text{OF}_C(\rightsquigarrow)$ .*

*Proof.* Using theorem 12, it is sufficient to prove that for any  $C$ , considering any  $\rightsquigarrow \notin D$  (thus,  $\text{OF}_C(\rightsquigarrow)$ ),  $\text{OF}_{D \setminus C}(\rightsquigarrow)$  holds. This is given by theorem 15.  $\square$

Given  $\rightsquigarrow, \rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , say  $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow'$  iff  $\rightsquigarrow$  and  $\rightsquigarrow'$  differ at only finitely many pairs  $(i, \alpha)$ , that is, iff  $\#\{(i, \alpha) \in I \times \mathcal{A}^{<\mathbb{N}} \mid \rightsquigarrow(i, \alpha) \neq \rightsquigarrow'(i, \alpha)\} \in \mathbb{N}$ .  $\sim_{\text{fin}}$  is an equivalence relation on  $P^{I \times \mathcal{A}^{<\mathbb{N}}}$ : reflexivity and symmetry are immediate, and transitivity holds because, if  $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow' \sim_{\text{fin}} \rightsquigarrow''$ , the pairs where  $\rightsquigarrow$  and  $\rightsquigarrow''$  differ are among those where  $\rightsquigarrow$  differs from  $\rightsquigarrow'$  or  $\rightsquigarrow'$  differs from  $\rightsquigarrow''$ , a finite union of two finite sets.

**Definition 2** (Closed under finite modification). A claim  $C$  is *closed under finite modification* iff  $\forall \rightsquigarrow \in C, \rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}} : \rightsquigarrow \sim_{\text{fin}} \rightsquigarrow' \Rightarrow \rightsquigarrow' \in C$ .

Being closed under finite modification and non-empty does not entail triviality.

*Example 14* (Closed under finite modification, yet non-trivial). Fix any  $\rightsquigarrow^{(0)} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$  and let  $C = \{\rightsquigarrow \mid \rightsquigarrow \sim_{\text{fin}} \rightsquigarrow^{(0)}\}$ , the  $\sim_{\text{fin}}$ -equivalence class of  $\rightsquigarrow^{(0)}$ .  $C$  is closed under finite modification, by transitivity of  $\sim_{\text{fin}}$ , and  $\rightsquigarrow^{(0)} \in C$ , yet  $C \neq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  (the world differing from  $\rightsquigarrow^{(0)}$  at every pair, say, is excluded).  $\triangle$

**Theorem 17.** *If  $C \neq \emptyset$  is closed under finite modification, then  $C$  is never falsifiable.*

*Proof.* Fix  $\rightsquigarrow^{(0)} \in C$  and let us show that  $\Theta \subseteq C^\subseteq$ . Given any  $\theta \in \Theta$ , let  $\rightsquigarrow$  agree with  $\theta$  on  $\theta(P)$  and with  $\rightsquigarrow^{(0)}$  elsewhere; then  $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow^{(0)}$ , as they differ only on the finite set  $\theta(P)$ , so  $\rightsquigarrow \in C$ , and  $\theta \subseteq \rightsquigarrow$ , giving  $\theta \in C^\subseteq$ .  $\square$

*Remark 10.* The converse of theorem 17 fails: example 6 is never falsifiable ( $\Theta \subseteq C^\subseteq$ ) yet is not closed under finite modification, as flipping a single individual's two values (making it satisfy neither disjunct) already leaves  $C$ .  $\triangle$

## A.6. POF

*Remark 11.* By corollary 5,  $\rightsquigarrow \in C \cap D$  satisfies established verifiability but not PV under  $D$  iff  $\text{OF}_{D \setminus C}(\rightsquigarrow) \wedge \neg \text{POF}_{D \setminus C}(\rightsquigarrow)$ , that is, iff  $D \setminus C$  suffers visibility issues at  $\rightsquigarrow$ . So, for a given claim and axiom, the gap between established verifiability and PV is precisely governed by visibility issues of  $D \setminus C$ : no such gap arises wherever  $D \setminus C$  suffers no visibility issues.  $\triangle$

A claim  $C$  is persistently falsifiable or trivial iff it is provable by  $\neg \text{POF}_C$ .

**Theorem 18.** A claim  $C$  is provable by  $\neg \text{POF}_C$  iff  $C = \emptyset \vee C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ .

*Proof.* If  $C = \emptyset$ , then  $\forall \theta \in \Theta : \theta \notin C^\subseteq$ , thus  $\forall \rightsquigarrow : \text{POF}_C(\rightsquigarrow)$ .

Conversely, suppose  $P^{I \times \mathcal{A}^{<\mathbb{N}}} \neq C \neq \emptyset$ ; we show  $\exists \rightsquigarrow \notin C \mid \neg \text{POF}_C(\rightsquigarrow)$ .

If  $C$  is closed under finite modification (definition 2), by theorem 17,  $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$ , thus  $\neg \text{POF}_C(\rightsquigarrow)$ ; suffices to pick any  $\rightsquigarrow \notin C$ .

If  $C$  is not closed under finite modification, for some  $n \in \mathbb{N}$ , some two worlds  $\rightsquigarrow^{(0)} \in C, \rightsquigarrow^{(n)} \notin C$  differ at exactly  $n$  pairs. We can build a sequence of worlds  $(\rightsquigarrow^{(k)})_{1 \leq k \leq n-1}$ , by flipping one pair at a time, such that  $\forall 1 \leq k \leq n$ , each  $\rightsquigarrow^{(k)}$  differs from  $\rightsquigarrow^{(k-1)}$  at exactly one pair. Taking  $j = \min\{1 \leq k \leq n \mid \rightsquigarrow^{(k)} \notin C\}$ , we obtain worlds  $\rightsquigarrow^{(j-1)} \in C, \rightsquigarrow^{(j)} \notin C$  that differ at exactly one pair  $(i, \alpha)$ . To show  $\neg \text{POF}_C(\rightsquigarrow^{(j)})$ , pick any  $\alpha'$  satisfying  $\#\alpha < \#\alpha'$ , define  $\theta = \{(i, \alpha', \rightsquigarrow^{(j)}(i, \alpha'))\}$ , pick any  $\theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(j)})$ , and let us show that  $\theta' \in C^\subseteq$ . Indeed,  $(i, \alpha, \cdot) \notin \theta'$ , because  $\theta'$  observing  $i$  must use only extensions of  $\alpha'$ , itself longer than  $\alpha$ ; thus,  $\theta' \subseteq \rightsquigarrow^{(j-1)}$ , as  $\rightsquigarrow^{(j-1)}$  agrees with  $\rightsquigarrow^{(j)}$  on every other pair than  $(i, \alpha)$ .  $\square$

**Theorem 19.** A claim  $C$  is PV under  $D = \neg \text{POF}_C$  iff  $C = \emptyset \vee C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ .

*Proof.* If  $C = \emptyset$ , then  $\forall \theta \in \Theta : \theta \notin C^\subseteq$ , thus  $\forall \rightsquigarrow : \text{POF}_C(\rightsquigarrow)$ .

Conversely, suppose  $P^{I \times \mathcal{A}^{<\mathbb{N}}} \neq C \neq \emptyset$ ; we show  $\exists \rightsquigarrow \in C \mid \neg \text{POF}_C(\rightsquigarrow)$ .

PV demands that all worlds in  $C$  satisfy PV at  $\rightsquigarrow$ :  $\forall \theta, \exists \theta \sqsubseteq \theta' \notin (D \setminus C)^\subseteq$ . In other words, that  $\text{POF}_{D \setminus C}(\rightsquigarrow)$ .

We show  $\exists \rightsquigarrow \in C \mid$  for some prefix  $\theta$ , all completions  $\theta'$  are completable in  $D \setminus C$ .

If  $C$  is closed under finite modification (definition 2), by theorem 17,  $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$ , thus  $\neg \text{POF}_C(\rightsquigarrow)$ , thus  $D \setminus C = \overline{C}$ . As  $\sim_{\text{fin}}$  is an equivalence relation,  $C$  is closed under finite modification iff  $\overline{C}$  is closed under finite modification. Therefore, by theorem 17 again,  $\forall \rightsquigarrow : \neg \text{POF}_{\overline{C}}(\rightsquigarrow)$ .

If  $C$  is not closed under finite modification, for some  $n \in \mathbb{N}$ , some two worlds  $\rightsquigarrow^{(0)} \in C, \rightsquigarrow^{(n)} \notin C$  differ at exactly  $n$  pairs. We can build a sequence of worlds  $(\rightsquigarrow^{(k)})_{1 \leq k \leq n-1}$ , by flipping one pair at a time, such that  $\forall 1 \leq k \leq n$ , each  $\rightsquigarrow^{(k)}$  differs from  $\rightsquigarrow^{(k-1)}$  at exactly one pair. Taking  $j = \min\{1 \leq k \leq n \mid \rightsquigarrow^{(k)} \notin C\}$ , we obtain worlds  $\rightsquigarrow^{(j-1)} \in C, \rightsquigarrow^{(j)} \notin C$  that differ at exactly one pair  $(i, \alpha)$ . Pick any  $\alpha'$  satisfying  $\#\alpha < \#\alpha'$ , define  $\theta = \{(i, \alpha', \rightsquigarrow^{(j-1)}(i, \alpha'))\}$ , pick any  $\theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(j-1)})$ , and let us show that  $\theta' \in (D \setminus C)^\subseteq$ . Indeed,  $(i, \alpha, \cdot) \notin \theta'$ , because  $\theta'$  observing  $i$  must use only extensions of  $\alpha'$ , itself longer than  $\alpha$ . Thus,  $\theta' \subseteq \rightsquigarrow^{(j)}$ , as  $\rightsquigarrow^{(j)}$  agrees with  $\rightsquigarrow^{(j-1)}$  on every other pair than  $(i, \alpha)$ . Furthermore,  $\neg \text{POF}_C(\rightsquigarrow^{(j)})$  because  $\theta \subseteq \rightsquigarrow^{(j)}$  and  $\theta' \subseteq \rightsquigarrow^{(j-1)} \in C$ .  $\square$

*Remark 12.* I believe that we're basically finding a prefix  $\theta$  such that  $\forall \theta' \sqsubseteq \theta : \theta' \subseteq \rightsquigarrow^{(j-1)} \in C \wedge \theta' \subseteq \rightsquigarrow^{(j)} \notin C$ . Also, perhaps, existence of such a prefix is equivalent to the statement of the theorem. This may perhaps bring some better understanding. To be checked.  $\triangle$

## A.7. Observational completion

**Corollary 9.** *Given any claims  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , possible verifiability under axiom  $D$ , that is,  $\exists \theta \in \Theta \cap D^\subseteq \mid \forall \theta' \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$ , is equivalent to  $D$  being observationally completable to  $C$ , that is,  $\exists \theta \in \Theta \cap D^\subseteq \mid \theta \notin (D \setminus C)^\subseteq$ .*

*Remark 13.* Given any claims  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ ,  $[C$  satisfies established verifiability under axiom  $D$  ( $\forall \rightsquigarrow \in C \cap D, \text{OF}_{D \setminus C}(\rightsquigarrow)$ ) is equivalent to  $D$  is necessarily observationally completable to  $C$  ( $\forall \rightsquigarrow \in D, \text{OF}_{D \setminus C}(\rightsquigarrow)$ )] iff  $D \setminus C = \emptyset$ . Indeed, assume EV under  $D$  and consider  $\rightsquigarrow \in D \cap \overline{C}$ , we have  $\neg \text{F}_{D \setminus C}(\rightsquigarrow)$ , thus  $\neg \text{OF}_{D \setminus C}(\rightsquigarrow)$ .  $\triangle$

**Corollary 10.** *Given any claims  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , if some claim  $C' \subseteq C$  is possibly verifiable under axiom  $D$ , that is,  $\exists \theta \in \Theta \cap D^\subseteq \mid \forall \theta' \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C'$ , then  $D$  is observationally completable to  $C$ , that is,  $\exists \theta \in \Theta \cap D^\subseteq \mid \theta \notin (D \setminus C)^\subseteq$ .*

**Theorem 20.** *Given any  $E$  and a cover  $\mathcal{S}$  of  $E$  (a set of non-empty subsets whose union equals  $E$ ) such that  $\forall S \in \mathcal{S}, \exists f_S \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid S = \uparrow f_S$  with  $f_S$  right unique, given any  $f \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P$ :  $f \notin E^\subseteq$  is equivalent to  $\forall S \in \mathcal{S}, \exists t \in f \mid \{t\} \notin S^\subseteq$ .*

*Proof.* Pick any  $S \in \mathcal{S}$  and  $f_S \mid S = \uparrow f_S$ . Given any  $p \in f_S$ , define  $S_p = \uparrow \{p\}$ , obtaining  $S = \bigcap_{p \in f_S} S_p$ . Thus  $f \cup f_S$  is not completable to a function (if  $f \cup f_S$  is completable to a function, that function is in  $S$ , thus in  $E$ , thus  $f$  compatible with  $E$ ). It follows that  $f \cup f_S$  is not right unique, thus, that some  $t \in f$  contradicts  $f_S$ .

Conversely, if  $f \in E^\subseteq$ , then  $f \subseteq \rightsquigarrow$  for some  $\rightsquigarrow \in E$ . As  $\mathcal{S}$  covers  $E$ ,  $\rightsquigarrow \in S$  for some  $S \in \mathcal{S}$ . Then  $\forall t \in f$ :  $\{t\} \subseteq \rightsquigarrow \in S$ , thus  $\{t\} \in S^\subseteq$ . Thus, for this  $S$ , no  $t \in f$  satisfies  $\{t\} \notin S^\subseteq$ .  $\square$

*Remark 14.*  $(a \neq b) \vee (a \wedge c)$  is equivalent to  $(a \neq b) \vee (b \wedge c)$ , thus there is no unique minimal DNF.  $\triangle$

The following result indicates that, given a finite cover of  $D \setminus C$ ,  $\text{OF}_{D \setminus C}(\rightsquigarrow)$  requires some set of triple  $T$  that, for each  $S$ , contains a triple incompatible with  $S$ ; thus  $\text{OF}_{D \setminus C}(\rightsquigarrow)$  holds iff such a  $T$  is observable in the world.

**Theorem 21.** *Given  $C, D$ , define  $E = D \setminus C$ , consider a finite cover  $\mathcal{S}$  of  $E$  (cover is understood as above). The world  $\rightsquigarrow$  satisfies  $\text{OF}_E(\rightsquigarrow)$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall S \in \mathcal{S}, \exists t \in \theta \mid \{t\} \notin S^\subseteq$ .*

*Proof.* By definition,  $\text{OF}_E(\rightsquigarrow)$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin E^\subseteq$ . By theorem 20, applied to  $f = \theta$ ,  $\theta \notin E^\subseteq$  iff  $\forall S \in \mathcal{S}, \exists t \in \theta \mid \{t\} \notin S^\subseteq$ .  $\square$

## A.8. Repeatability

**Theorem 22.** *Given  $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ :  $(\neg \text{POF}_C) \cap D$  proves  $C$  iff  $(\neg \text{OF}_C) \cap D$  proves  $C$  and  $C$  suffers no visibility issues in  $D$  (that is,  $\forall \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$ ).*

*Proof.* The claim is equivalent to  $[\forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{POF}_C(\rightsquigarrow)] \Leftrightarrow [\forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \wedge \forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)]$ , because  $\forall \rightsquigarrow \in C, \neg \text{OF}_C(\rightsquigarrow)$ .  $\square$

*Example 15.* Define  $C = \alpha \rightsquigarrow_i \varphi$  for a given  $\alpha$  and  $i$ .

We have  $\neg \text{OF}_C \subseteq C$ , as visible by observing that  $\alpha \not\rightsquigarrow_i \varphi \Rightarrow \text{OF}_C(\rightsquigarrow)$ .

Define  $\rightsquigarrow$  such that  $\forall \alpha', j : \alpha' \rightsquigarrow_j \varphi'$  (thus  $\rightsquigarrow \in E$  and  $\rightsquigarrow \notin C$ ) and let us show that  $\neg \text{POF}_{C \cap E}(\rightsquigarrow)$ . Pick any  $\#\alpha < \#\alpha'$  and define  $\theta = (i, \alpha', \varphi') \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ . Consider any  $\theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ . Define an infinitely numerous subset  $J$  among the individuals for which  $\alpha$  was not observed in  $\theta'$  such that  $i \in J$ , leaving another infinitely numerous subset  $I \setminus J$ , and define  $\rightsquigarrow'$  such that  $\forall j \in J : \alpha \rightsquigarrow'_j \varphi \wedge \forall \alpha'' \neq \alpha, \alpha'' \rightsquigarrow'_j \varphi'$ , and  $\forall j \in I \setminus J, \alpha'' : \alpha'' \rightsquigarrow'_j \varphi'$ . We have  $\theta' \subseteq \rightsquigarrow' \in C \cap E$ , thus,  $\theta' \in (C \cap E)^\subseteq$ .  $\triangle$

Example 15 shows that a claim can be provable by  $\neg \text{OF}_C$  (thus, provable by  $\neg \text{OF}_{C \cap E} \cap E$ ) yet not provable by  $\neg \text{POF}_{C \cap E} \cap E$  (thus, also not provable by  $\neg \text{POF}_C \cap E$ ). (Recall that  $\neg \text{POF}_{C \cap E} \subseteq \neg \text{POF}_C$  and  $\neg \text{OF}_{C \cap E} \subseteq \neg \text{OF}_C$ .)

Say that a world admits a proof of  $\neg C$  avoiding  $J$  iff  $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$ . A sufficient condition for a world being POF is that for any finite subset of individuals, some proof of  $\neg C$  avoids that subset.

**Theorem 23.** *Given any claim  $C$  and world  $\rightsquigarrow$ :  $[\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq] \Rightarrow \text{POF}_C(\rightsquigarrow)$ .*

*Claude proof.* Fix  $\tau \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ; let  $J \subseteq I$  be its (finite) individuals, and pick, by hypothesis,  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ,  $\theta \notin C^\subseteq$ , touching only individuals outside  $J$ . As  $\tau, \theta$  touch disjoint individuals,  $\tau' = \tau \cup \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  and  $\tau \sqsubseteq \tau'$  (no point is added for any individual already in  $\tau$ 's domain). As  $C^\subseteq$  is downwards closed and  $\theta \subseteq \tau'$ ,  $\tau' \notin C^\subseteq$ . As  $\tau$  was arbitrary,  $\text{POF}_C(\rightsquigarrow)$  holds.  $\square$

The converse of theorem 23 fails: when some given  $i$  is non avoidable for a proof of  $\neg C$  but all  $\alpha$  do prove  $\neg C$ , it does not bar POF.

*Example 16.* Fix  $i^* \in I$ . Let  $\rightsquigarrow(i, \alpha) = \varphi$  for all  $i \in I$ ,  $\alpha \in \mathcal{A}^{<\mathbb{N}}$ , and  $C = \{v \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \alpha \in \mathcal{A}^{<\mathbb{N}} : v(i^*, \alpha) = \varphi'\}$ . Then  $\rightsquigarrow \notin C$  and  $\text{POF}_C(\rightsquigarrow)$ : given any  $\tau \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ , if  $i^*$  is not queried by  $\tau$ , extend  $\tau$  by one observation of  $i^*$  at  $\emptyset$  to get  $\tau'$  with  $\tau \sqsubseteq \tau'$  and  $\tau' \notin C^\subseteq$  (as  $\rightsquigarrow(i^*, \emptyset) = \varphi \neq \varphi'$ ); if  $i^*$  is already queried,  $\tau$  records  $\varphi$  at some path, so no  $v \in C$  (requiring  $\varphi'$  everywhere for  $i^*$ ) extends  $\tau$ , and  $\tau' = \tau$  works. Yet poofsoi fails for  $J = \{i^*\}$ : any  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  querying only in  $I \setminus \{i^*\}$  lies in  $C^\subseteq$ , since setting  $v(i^*, \alpha) = \varphi'$  for all  $\alpha$  and  $v(i, \alpha) = \varphi$  for  $i \neq i^*$  yields  $v \in C$  with  $\theta \subseteq v$ .  $\triangle$

A world  $\rightsquigarrow$  differs for infinitely many individuals with  $E$  iff it differs for infinitely many individuals with each world in  $E$ ; formally, iff  $\forall \rightsquigarrow \neq \rightsquigarrow' \in E : \#\{i \in I \mid \rightsquigarrow_i \neq \rightsquigarrow'_i\} \notin \mathbb{N}$ .

*Example 17* (A world differing for infinitely many individuals with  $E$ ). Let  $w^*(i, \alpha) = \varphi'$  for all  $i \in I$ ,  $\alpha \in \mathcal{A}^{<\mathbb{N}}$  (constant world). Then  $w^* \in E$  (all individuals share the same behavior). For each  $\rightsquigarrow \in E \setminus \{w^*\}$ :  $\rightsquigarrow \neq w^*$  means some individual  $j$  has  $\rightsquigarrow(j, \cdot) \neq$  (always  $\varphi'$ ); since  $\rightsquigarrow \in E$ , this behavior  $\rightsquigarrow(j, \cdot)$  is shared by infinitely many individuals  $i$ , each satisfying  $\rightsquigarrow(i, \cdot) \neq$  (always  $\varphi'$ )  $= w^*(i, \cdot)$ . Hence  $\#\{i \in I \mid \rightsquigarrow(i, \cdot) \neq w^*(i, \cdot)\} \notin \mathbb{N}$  for every  $\rightsquigarrow \in E \setminus \{w^*\}$ :  $w^*$  differs for infinitely many individuals with  $E$ .  $\triangle$

*Example 18* (A world not differing for infinitely many individuals with  $E$ ). Fix  $a \in \mathcal{A}$ , an infinite proper subset  $S \subsetneq I$  with  $I \setminus S$  infinite, and  $i_0 \in S$ . Let  $v \in E$  be the world where  $v(i, (a)) = \varphi'$  for  $i \in S$  and  $v(i, \alpha) = \varphi$  otherwise. Let  $w_0$  agree with  $v$  everywhere except  $w_0(i_0, (a)) = \varphi$ . Then  $w_0 \in E$ : individuals in  $S \setminus \{i_0\}$  all share behavior " $\varphi'$  at  $(a)$ ,  $\varphi$  elsewhere" (infinitely many), and individuals in  $\{i_0\} \cup (I \setminus S)$  all share behavior "always  $\varphi$ " (infinitely many). The worlds  $w_0$  and  $v$  differ only at  $i_0$ , so  $\#\{i \in I \mid w_0(i, \cdot) \neq v(i, \cdot)\} = 1 \in \mathbb{N}$ :  $w_0$  does not differ for infinitely many individuals with  $v \in E$ .  $\triangle$

**Lemma 2.** *Given any anonymous  $C$  and any world  $\rightsquigarrow \in E$  such that  $\text{OF}_C(\rightsquigarrow)$ :  $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$ .*

*Proof.* As  $\text{OF}_C(\rightsquigarrow)$ , some  $\theta^{(0)} \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  satisfies  $\theta^{(0)} \notin C^\subseteq$ . Define  $J^{(0)} = \theta^{(0)}(\mathcal{A}^{<\mathbb{N}}, P)$  as the individuals queried by  $\theta^{(0)}$ .

Given any  $J \subseteq_{\text{fin}} I$ , define  $r$  as an injection from  $J^{(0)}$  to  $I \setminus (J \cup J^{(0)})$  such that  $\rightsquigarrow_{r(i)} = \rightsquigarrow_i$  (which exists because  $\rightsquigarrow \in E$ ), define  $J' = r(J^{(0)})$  (thus  $r$  is a bijection from  $J^{(0)}$  to  $J'$  disjoint from both  $J$  and  $J^{(0)}$ ), complete  $r$  to the self-inverse permutation  $\sigma = r \cup r^{-1} \cup \{(i, i) \mid i \notin J^{(0)} \cup J'\}$  over  $I$ , and define  $\theta = \{(r(i), \alpha, p) \mid (i, \alpha, p) \in \theta^{(0)}\}$  as  $\theta^{(0)}$  where individuals from  $J^{(0)}$  are replaced by individuals determined by  $r$ . We observe that  $\theta \subseteq \rightsquigarrow$  (by construction of  $r$ ) and that  $\theta(\mathcal{A}^{<\mathbb{N}}, P) = J' \subseteq I \setminus (J \cup J^{(0)}) \subseteq I \setminus J$ . Finally, if there exists  $\theta \subseteq \rightsquigarrow_J \in C$ , then also  $\theta^{(0)} \subseteq \rightsquigarrow_J \circ \sigma^{-1} \in C$  (by anonymity of  $C$ ); as  $\theta^{(0)} \notin C^\subseteq$ , we obtain  $\theta \notin C^\subseteq$ .  $\square$

**Lemma 3.** *Given any two claims  $C_1, C_2$  and world  $\rightsquigarrow$ : if  $C_1$  and  $C_2$  each satisfy  $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$  (with  $C = C_1$  and  $C = C_2$  respectively), then so does  $C_1 \cup C_2$ .*

*Proof.* Given any  $J \subseteq_{\text{fin}} I$ : apply the hypothesis for  $C_2$  to  $J$ , obtaining  $\theta_2 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  with  $\theta_2(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J$  and  $\theta_2 \notin C_2^\subseteq$ ; let  $J_2 = \theta_2(\mathcal{A}^{<\mathbb{N}}, P)$ . Apply the hypothesis for  $C_1$  to  $J \cup J_2$ , obtaining  $\theta_1$  with  $\theta_1(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus (J \cup J_2)$  and  $\theta_1 \notin C_1^\subseteq$ . Set  $\theta = \theta_1 \cup \theta_2$ : since  $\theta_1$  and  $\theta_2$  query disjoint individuals and are both consistent with  $\rightsquigarrow$ ,  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  and  $\theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J$ . If  $\theta \subseteq v$  for some  $v \in C_1$ , then  $\theta_1 \subseteq \theta \subseteq v$ , contradicting  $\theta_1 \notin C_1^\subseteq$ ; so  $\theta \notin C_1^\subseteq$ . Likewise  $\theta_2 \subseteq \theta$  implies  $\theta \notin C_2^\subseteq$ . Hence  $\theta \notin (C_1 \cup C_2)^\subseteq$ .  $\square$

**Lemma 4.** *Given any finite  $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  and world  $\rightsquigarrow$  such that  $\forall v \in F : \#\{i \in I \mid \rightsquigarrow_i \neq v_i\} \notin \mathbb{N}$ :  $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin F^\subseteq$ .*

*Proof.* Given any  $J \subseteq_{\text{fin}} I$ , define  $s$  as an injection from  $F$  to  $I \setminus J$  such that  $w_{s(w)} \neq \rightsquigarrow_i$ , and  $\alpha_{s(w)}$  a path such that  $w(\alpha_{s(w)}, s(w)) \neq \rightsquigarrow(\alpha_{s(w)}, s(w))$ . Define  $\theta = \{(s(w), \alpha_{s(w)}, \rightsquigarrow(\alpha_{s(w)}, s(w))) \mid w \in F\}$ .

Then  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ,  $\theta(\mathcal{A}^{<\mathbb{N}}, P) = \{i_v \mid v \in F\} \subseteq I \setminus J$ , and for each  $v' \in F$ :  $(\alpha_{v'}, i_{v'}, \rightsquigarrow(\alpha_{v'}, i_{v'})) \in \theta$  with  $\rightsquigarrow(\alpha_{v'}, i_{v'}) \neq v'(\alpha_{v'}, i_{v'})$ , so  $\theta \not\subseteq v'$ , hence  $\theta \notin F^\subseteq$ .  $\square$

**Lemma 5.** *Given any anonymous  $C_0$ , finite  $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  with  $\forall w \in F, \forall v \in E \setminus \{w\} : \#\{i \in I \mid v_i \neq w_i\} \notin \mathbb{N}$ ,  $C = C_0 \cup F$ , and any world  $\rightsquigarrow \in E$  such that  $\text{OF}_C(\rightsquigarrow) : \forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$ .*

*Proof.* As  $\text{OF}_C(\rightsquigarrow)$ ,  $\rightsquigarrow \notin C$ , so  $\rightsquigarrow \notin F$ ; as  $\rightsquigarrow \in E$ , the hypothesis on  $F$  gives  $\forall v \in F : \#\{i \in I \mid \rightsquigarrow_i \neq v_i\} \notin \mathbb{N}$ . Moreover, any  $\theta \notin C^\subseteq$  also lies outside  $C_0^\subseteq$  (as  $C_0 \subseteq C$  implies  $C_0^\subseteq \subseteq C^\subseteq$ ), so  $\text{OF}_{C_0}(\rightsquigarrow)$  holds. Apply lemma 2 to  $C_0$ , lemma 4 to  $F$ , and lemma 3:  $C = C_0 \cup F$  satisfies the conclusion.  $\square$

This follows from theorem 23 and lemma 5.

**Theorem 24** (Finitely many, individually-spread exceptions to an anonymous claim do not register in  $E$ ). *If  $C_0$  is anonymous,  $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  is finite, and  $\forall w \in F, \forall \rightsquigarrow \in E \setminus \{w\} : \#\{i \in I \mid \rightsquigarrow_i \neq w_i\} \notin \mathbb{N}$ , then  $C = C_0 \cup F$  suffers no visibility issues in  $E$ .*

*Example 19* (Instance: a single named exception, escaping  $E$ ). Let  $C_0 = \{\rightsquigarrow \mid \forall i \in I : \rightsquigarrow(i, a) = \varphi\}$  (anonymous), fix  $i^* \in I$ , and let  $\rightsquigarrow^{(0)}$  agree with  $\varphi$  everywhere except  $\rightsquigarrow^{(0)}(i^*, a) = \varphi'$ . By theorem 24 (with  $F = \{\rightsquigarrow^{(0)}\}$ ),  $C = C_0 \cup \{\rightsquigarrow^{(0)}\}$  suffers no visibility issues in  $E$ , yet is not anonymous (swapping  $i^*$  with any other individual moves the exception away from both  $C_0$  and  $\rightsquigarrow^{(0)}$ ). Notably,  $\rightsquigarrow^{(0)} \notin E$ : its one distinguished individual  $i^*$  has no clones.  $\triangle$

**Theorem 25.** *If an anonymous claim  $C \subseteq E$  is falsifiable or trivial, it is provable by  $\neg \text{POF}_C \cap E$ .*

*Proof.*  $C$  is falsifiable or trivial iff  $\neg \text{OF}_C \subseteq C$ , which implies  $\neg \text{OF}_C \cap E \subseteq C$ .

From theorem 24 (with  $F = \emptyset$ ),  $C$  suffers no visibility issues in  $E$ . From theorem 22,  $\neg \text{POF}_C \cap E$  proves  $C$  iff  $\neg \text{OF}_C \cap E$  proves  $C$ .  $\square$

The converse fails, as example 20 shows.

**Theorem 26.** *If  $\#I \notin \mathbb{N}$ , given any general observation  $O$ ,  $\exists O \subseteq \rightsquigarrow \in E$ .*

*Proof.* Define for each individual observed in  $O$  a behavior that completes the observation that concern her in  $O$ . Define  $\rightsquigarrow$  as the world associating infinitely many individuals to each of these behaviors.  $\square$

*Example 20.* The claim  $C = E$  is anonymous, provable by  $\neg \text{POF}_C \cap E$ , not trivial and not falsifiable: consider any  $\rightsquigarrow \notin E$  (such as a world associating a given behavior to a single individual  $i$  and a different behavior to all others), we have  $\text{F}_C(\rightsquigarrow)$  but  $\neg \text{OF}_C(\rightsquigarrow)$  as any observation  $\theta$  is completable in  $C = E$  (theorem 26).  $\triangle$

**Theorem 27.** *Given any  $\rightsquigarrow \in E$ , there exists an anonymous claim that is provable by  $\neg \text{POF}_{C \cap E} \cap E$ .*

*Proof.* Take  $C = E$ .  $E$  is anonymous: for any  $\rightsquigarrow \in E$  and permutation  $\sigma$ , individual  $j$  in  $\rightsquigarrow \circ \sigma$  has the same behavior as  $\sigma(j)$  in  $\rightsquigarrow$ , so the behavior classes of  $\rightsquigarrow \circ \sigma$  are those of  $\rightsquigarrow$  relabeled, each still infinite, giving  $\rightsquigarrow \circ \sigma \in E$ .  $\rightsquigarrow \in E = C$  by hypothesis.  $C \cap E = E$ , so  $(\neg \text{POF}_{C \cap E}) \cap E = (\neg \text{POF}_E) \cap E \subseteq E = C$ .  $\square$

**Theorem 28.** *Given any  $\rightsquigarrow \in E$ , there exists an anonymous claim that holds, is informative and provable by  $\neg \text{POF}_{C \cap E} \cap E$ .*

*Proof.* Take  $C = E$ .  $E$  is anonymous (as in the previous proof),  $\rightsquigarrow \in E = C$ , and  $(\neg \text{POF}_{C \cap E}) \cap E = (\neg \text{POF}_E) \cap E \subseteq E = C$ . *Informativeness.* For any  $\rightsquigarrow \in E$ , each behavior class is infinite and individuals sharing a behavior share the same  $\pi$ -value, so every non-empty fiber of  $\pi(\rightsquigarrow)$  is infinite. Hence, fixing any  $i_0 \in I$  and  $v \in P$ , the profile assigning  $v$  to  $i_0$  and  $\emptyset$  to all other individuals has a  $v$ -fiber of size  $1 \in \mathbb{N}$ , so it lies outside  $\pi(E)$ , giving  $\pi(E) \neq (P \cup \{\emptyset\})^I$ .  $\square$

In some worlds, the axiom  $\neg \text{POF}_{C \cap E} \cap E$  is sufficient to get existence of suitable claims.

*Example 21.* Consider the world  $\rightsquigarrow = J \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi\} \cup K \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi'\}$  of example 2, with infinite sets of individuals  $J, K$ . The claim  $C = [(\forall i : \emptyset \rightsquigarrow_i \varphi \Rightarrow \forall \alpha, \alpha \rightsquigarrow_i \varphi) \wedge (\exists i \mid \emptyset \rightsquigarrow_i \varphi)] \cap E$  holds, is anonymous, APTS and PEF under  $\neg \text{POF}_{C \cap E} \cap E$ .  $\triangle$

We are here interested in claims implying propositions of the form: any individual whose behavior satisfies some given observable condition admits a given influence being decisive. Given  $T : H \rightarrow P$  for some finite  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ , write  $T(i) = \{i\} \times T = \{(i, \alpha, p) \mid (\alpha, p) \in T\} \in \Theta$  for the observation that  $i$  conforms to  $T$  on the increasing path  $H$ .

Given  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  and  $H \subseteq \mathcal{A}^{<\mathbb{N}}$ , let  $B|_H = \{f|_H \mid f \in B\}$  denote the behaviors in  $B$  restricted to  $H$ . Say a behavior  $g$  is undistinguishable from  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  iff  $\forall H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) : g|_H \in B|_H$ . The claim  $C_B$  associated to  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  is the claim  $C_B = [\rightsquigarrow(I) \subseteq B]$ .

**Theorem 29.** *Consider  $\rightsquigarrow \in E$ , an anonymous claim  $C$  that holds, and a world  $\rightsquigarrow'$ . If  $\forall b \in \rightsquigarrow'(I)$ ,  $b$  is undistinguishable from  $\rightsquigarrow(I)$ , then  $\neg \text{OF}_C(\rightsquigarrow')$ .*

*Proof.* Consider any  $\theta \subseteq \rightsquigarrow'$  and let us show that  $\theta \in C^\subseteq$ .

Define the injection  $s : \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow I$  as follows. Consider each  $i \in \theta(\mathcal{A}^{<\mathbb{N}}, P)$  in turn. Let  $H = \theta(i, P) \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ . As  $\rightsquigarrow'_i$  is undistinguishable from  $\rightsquigarrow(I)$ ,  $\exists i' \mid \rightsquigarrow_{i'}|_H = \rightsquigarrow_i|_H$ . Set  $s(i) = i'$ . As  $\rightsquigarrow \in E$ , there are infinitely many suitable  $i'$  so we can pick one that has not been chosen yet, to ensure  $s$  is an injection.

The function  $s$  has domain  $\theta(\mathcal{A}^{<\mathbb{N}}, P)$  and codomain  $s(\theta(\mathcal{A}^{<\mathbb{N}}, P))$ , both finite sets of equal cardinality. Turn  $s$  into a permutation  $\sigma = s \cup s'$  over  $I$  by defining any bijection  $s' : I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow I \setminus s(\theta(\mathcal{A}^{<\mathbb{N}}, P))$  (TBD: argue that we can do this).

By construction,  $\theta \subseteq \rightsquigarrow \circ \sigma$ , which belongs to  $C$  because  $C$  holds and is anonymous.  $\square$

**Theorem 30.** *Given  $\rightsquigarrow \in E$ ,  $a \in \mathcal{A}$ ,  $b^{new} \in P^{\mathcal{A}^{<\mathbb{N}}}$ , some finite  $\emptyset \neq H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  and an anonymous claim  $C \subseteq [\forall i \in I : b^{new}|_H \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$ , if  $b^{new}$  is undistinguishable from  $\rightsquigarrow(I)$  yet  $b^{new} \notin \rightsquigarrow(I)$  and  $\nexists p \in P \mid a \hookrightarrow_{b^{new}} p$  and  $C$  holds, it is not verifiable at  $\rightsquigarrow$  under  $\neg \text{OF}_C \cap E$  (thus also not verifiable at  $\rightsquigarrow$  under  $\neg \text{POF}_C \cap E$ ).*

*Proof.* Consider any prefix  $\theta \subseteq \rightsquigarrow$ . We build  $\theta \subseteq \rightsquigarrow' \in (\neg \text{OF}_C \cap E) \setminus C$ . Consider any  $i \in \theta(\mathcal{A}^{<\mathbb{N}}, P)$ . Pick any  $b \in \rightsquigarrow(I) \mid \theta(i) \subseteq b$ , define  $\rightsquigarrow_{i'}' = b$ . Partition  $I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$  into infinite sets, one for each behavior in  $\rightsquigarrow(I)$ , plus one for  $b^{\text{new}}$ , thereby creating a function  $f : I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow P^{\mathcal{A}^{<\mathbb{N}}}$  (TBD argue from  $\#\rightsquigarrow(I) \leq \#I$  that we can do this). For each  $i \in I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$  with  $f(i) \in \rightsquigarrow(I)$ , define  $\rightsquigarrow_{i'}' = f(i)$ , for the remaining infinite set, define  $\rightsquigarrow_{i'}' = b^{\text{new}}$ .

Note that  $\rightsquigarrow' \in E$ . Also,  $\rightsquigarrow' \notin C$ : picking any  $i \mid \rightsquigarrow_{i'}' = b^{\text{new}}$ ,  $b^{\text{new}}|_H \subseteq \rightsquigarrow_{i'}'$ , but  $\nexists p \mid a \hookrightarrow_i' p$ .

Observe that each behavior in  $\rightsquigarrow'(I)$  is undistinguishable from  $\rightsquigarrow(I)$ . By theorem 29,  $\neg \text{OF}_C(\rightsquigarrow')$ .  $\square$

**Conjecture 1.** Consider  $\rightsquigarrow \in E$  such that for no  $(\mathbf{g}, a, p)$  with  $\exists i \in I : \mathbf{g} \subseteq \rightsquigarrow_i : \forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ . Then no anonymous, conclusive claim that holds is PV under  $(\neg \text{OF}_C) \cap E$ .

*Proof attempt.* Write  $D := (\neg \text{OF}_C) \cap E$  and suppose, towards a contradiction, that some anonymous, conclusive claim  $C$  holds and is PV under  $D$ .

*Step 1.* As  $C$  holds,  $\rightsquigarrow \in C$ ; since  $\rightsquigarrow$  itself extends every finite prefix  $\theta \subseteq \rightsquigarrow$  and lies in  $C$ , every such  $\theta$  is in  $C^\subseteq$ , so  $\neg \text{OF}_C(\rightsquigarrow)$ ; with  $\rightsquigarrow \in E$ ,  $\rightsquigarrow \in C \cap D$ .

*Step 2.* As  $C$  is conclusive and  $\rightsquigarrow \in C$ , applying conclusiveness at  $\theta = \emptyset$  gives  $\theta_0 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ,  $i^* \in I$ ,  $p^* \in P$  with  $\pi_{i^*}(\uparrow \theta_0 \cap C) = \{p^*\}$ . By corollary 5 and PV,  $\text{POF}_{D \setminus C}(\rightsquigarrow)$ ; applying at  $\theta_0$  gives  $\theta_0 \sqsubseteq \theta_1$  in  $\Theta \cap \mathcal{P}(\rightsquigarrow)$  with  $\uparrow \theta_1 \cap D \subseteq C$ , hence:

$$\forall \rightsquigarrow \in \uparrow \theta_1 \cap D : \pi_{i^*}(\rightsquigarrow) = p^*. \quad (\star)$$

Let  $\mathbf{g} := \theta_1|_{i^*}$  (so  $\mathbf{g} \subseteq \rightsquigarrow_{i^*}$  and  $\exists i : \mathbf{g} \subseteq \rightsquigarrow_i$ ).

*Step 3.* For any  $a \in \mathcal{A}$ : if every world in  $C$  had every  $\mathbf{g}$ -satisfying individual  $a$ -decisive for  $p^*$ , then  $\rightsquigarrow \in C$  would witness the specific-arg pattern  $(\mathbf{g}, a, p^*)$  in  $\rightsquigarrow$ , contradicting the hypothesis. Hence for each  $a$  there exists a world in  $C$  where some  $\mathbf{g}$ -satisfying individual is  $a$ -non-decisive.

Let  $b_0 := \rightsquigarrow_{i^*}$  (satisfies  $\mathbf{g}$ ). For each  $a \in \mathcal{A}$ : pick  $\alpha_a \notin \mathbf{g}^{-1}(P)$  with  $a \in \alpha_{\text{end}}(\alpha_a)$ , all  $\alpha_a$  distinct (possible since  $\mathbf{g}^{-1}(P)$  is finite). Define  $g'$  by  $g' = b_0$  on  $\mathbf{g}^{-1}(P)$ ,  $g'(\alpha_a) = p' \neq p^*$  for each  $a$ , and  $g' = b_0$  elsewhere. Then  $g'$  satisfies  $\mathbf{g}$  and has no decisive arg for  $p^*$ : for each  $a$ , the path  $\alpha_a$  witnesses  $a \not\hookrightarrow_{g'} p^*$ .

*Step 4.* Let  $w$  be the world obtained from  $\rightsquigarrow$  by replacing  $i^*$ 's behavior with  $g'$  (adjusting other individuals as needed to maintain  $w \in E$ ). Then  $w \in \uparrow \theta_1$  since  $\mathbf{g} = \theta_1|_{i^*} \subseteq g'$  and other individuals are unchanged.

For  $w \in \neg \text{OF}_C$ : any finite prefix  $\theta$  of  $w$  records  $i^*$ 's behavior on a finite chain of paths, witnessing  $a$ -non-decisiveness for only finitely many  $a$ 's. Since  $\mathcal{A}$  is infinite, some  $a_0$  has  $\alpha_{a_0}$  outside this chain; extending  $i^*$ 's behavior to be  $a_0$ -decisive (replacing  $p'$  by  $p^*$  at  $a_0$ -paths not yet observed) yields a world in  $C^\subseteq$ , so  $\theta \in C^\subseteq$ . Hence  $w \in \neg \text{OF}_C$ , giving  $w \in D$ .

With  $w \in \uparrow \theta_1 \cap D$ ,  $(\star)$  gives  $\pi_{i^*}(w) = p^*$ . But  $i^*$ 's behavior in  $w$  is  $g'$ , which has no decisive arg for  $p^*$ , so  $\pi_{i^*}(w) \neq p^*$ : a contradiction.  $\square$

**Theorem 31.** Fix  $\varphi \neq \varphi' \in P$  and  $a \neq b \in \mathcal{A}$ . Suppose  $\rightsquigarrow \in E$  contains behaviors  $(b_k)_{k \in \mathbb{N}}$ , each shared by infinitely many individuals in  $I$ , such that  $a \hookrightarrow_{b_0} \varphi$ ,  $b \not\hookrightarrow_{b_0} \varphi$ , and for each  $k \geq 1$ :  $b \hookrightarrow_{b_k} \varphi$ ,  $a \not\hookrightarrow_{b_k} \varphi$ , and  $b_k$  agrees with  $b_0$  on all paths of length  $\leq k$ . Then the claim  $C = [\forall i : a \hookrightarrow_i \varphi \vee b \hookrightarrow_i \varphi]$  is not PV under  $(\neg \text{POF}_C) \cap E$ .

*Proof.* Write  $D := (\neg \text{POF}_C) \cap E$ . By corollary 5, it suffices to exhibit  $v \in C \cap D$  with  $\neg \text{POF}_{D \setminus C}(v)$ . Take  $v := \rightsquigarrow$ : it holds ( $\rightsquigarrow \in C$ ), lies in  $E$  by hypothesis, and since  $\rightsquigarrow \in C$ , every  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  lies in  $C^\subseteq$  (witnessed by  $\rightsquigarrow$  itself), so  $\neg \text{POF}_C(\rightsquigarrow)$ : thus  $\rightsquigarrow \in C \cap D$ .

For  $\neg \text{POF}_{D \setminus C}(\rightsquigarrow)$ , fix  $\theta = \emptyset$  and let  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  have max depth  $m$ . Pick two paths  $\alpha_a, \alpha_b$  of length  $m + 1$  sharing the same length- $m$  prefix, with  $a \in \alpha_{\text{end}}(\alpha_a)$  and  $b \in \alpha_{\text{end}}(\alpha_b)$ ; they are incompatible (neither is a prefix of the other). Define behavior  $g'$  to agree with  $b_0$  on paths of length  $\leq m$ , set  $g'(\alpha_a) = g'(\alpha_b) = \varphi'$ , and  $g'(\alpha) = \varphi$  elsewhere. Then  $a \not\hookrightarrow_{g'} \varphi$  (witnessed by  $\alpha_a$ ) and  $b \not\hookrightarrow_{g'} \varphi$  (witnessed by  $\alpha_b$ ). Build  $w$  agreeing with  $\theta'$  on its support, giving  $g'$  to infinitely many fresh individuals, and distributing the rest among  $b_0, b_1, b_2, \dots$  (each to infinitely many): then  $w \in E$  and  $w \notin C$ .

It remains to verify  $w \in \neg \text{POF}_C$ , i.e., that every  $\theta \in \Theta \cap \mathcal{P}(w)$  lies in  $C^\subseteq$ . For any  $g'$ -individual  $j$ ,  $\theta$ 's observations of  $j$  form a chain, which can contain  $\alpha_a$  or  $\alpha_b$  but not both (they are incompatible). If  $\alpha_a \notin \theta$ : extend  $j$ 's behavior to be  $a$ -decisive (consistent with  $\theta$ , as no observed path ending in  $a$  gives  $\varphi'$ ), yielding a world in  $C$ . If  $\alpha_b \notin \theta$ : extend to a  $b$ -decisive behavior similarly. Either way  $\theta \in C^\subseteq$ , so  $w \in \neg \text{OF}_C \subseteq \neg \text{POF}_C$ , hence  $w \in D \setminus C$ .  $\square$

**Theorem 32.** Consider  $\rightsquigarrow \in E$  such that for no  $\mathbf{g}, p \in P$ :  $\forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow p \in P_i$ . Let  $C \mapsto F_C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  assign to every claim  $C$  some  $F_C$  with  $C \subseteq F_C$  that is anonymous ( $F_C \circ \sigma = F_C$  for every permutation  $\sigma$  of  $I$ ). Then, no anonymous, conclusive claim that holds is PV under  $(\neg \text{OF}_C) \cap E \cap F_C$ .

*Proof.* Write  $D := (\neg \text{OF}_C) \cap E \cap F_C$  and suppose, towards a contradiction, that some anonymous, conclusive claim  $C$  holds and is PV under  $D$ .

*Three invariance facts.* First,  $E$  is anonymous (as already observed when showing  $E$  itself is an anonymous claim provable by  $\neg \text{POF}_E \cap E$ ): relabelling individuals by a permutation  $\sigma$  permutes each behavior-class without changing its cardinality, so  $\forall \rightsquigarrow, \sigma : \rightsquigarrow \in E \Leftrightarrow \rightsquigarrow \circ \sigma \in E$ . Second, since  $C$  is anonymous,  $\text{OF}_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow \circ \sigma)$  for every  $\rightsquigarrow$  and every permutation  $\sigma$  of  $I$  (and likewise for  $\text{POF}_C$ ). Indeed, extending  $\rightsquigarrow \circ \sigma$ 's notation to (possibly partial) observations by  $\theta \circ \sigma := \{(i, \alpha, p) \mid (\sigma(i), \alpha, p) \in \theta\}$ , one checks directly from the definitions that  $\theta \subseteq \rightsquigarrow \Leftrightarrow \theta \circ \sigma \subseteq \rightsquigarrow \circ \sigma$  and  $\theta \sqsubseteq \theta' \Leftrightarrow \theta \circ \sigma \sqsubseteq \theta' \circ \sigma$ ; and, using  $C \circ \sigma = C$ : if  $\theta \subseteq v \in C$  then  $\theta \circ \sigma \subseteq v \circ \sigma \in C \circ \sigma = C$ , and conversely (applying  $\sigma^{-1}$  in place of  $\sigma$ ), so  $\theta \in C^\subseteq \Leftrightarrow \theta \circ \sigma \in C^\subseteq$ . Hence  $\theta \mapsto \theta \circ \sigma$  is a  $\sqsubseteq$ -preserving bijection between  $\Theta \cap \mathcal{P}(\rightsquigarrow)$  and  $\Theta \cap \mathcal{P}(\rightsquigarrow \circ \sigma)$  carrying  $\{\theta \notin C^\subseteq\}$  onto  $\{\theta \circ \sigma \notin C^\subseteq\}$ : this already gives  $\text{OF}_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow \circ \sigma)$ , since it matches nonempty witness sets to nonempty witness sets; and, since the bijection is also  $\sqsubseteq$ -preserving, it further carries any witness of  $\text{POF}_C(\rightsquigarrow)$  at some  $\theta$  to a witness of  $\text{POF}_C(\rightsquigarrow \circ \sigma)$  at  $\theta \circ \sigma$ , and conversely. Third,  $F_C \circ \sigma = F_C$  by hypothesis:  $\rightsquigarrow \in F_C \Leftrightarrow \rightsquigarrow \circ \sigma \in F_C$ .

*Step 1.* As  $C$  holds,  $\rightsquigarrow \in C$ , so  $\neg F_C(\rightsquigarrow)$ , hence  $\neg \text{OF}_C(\rightsquigarrow)$ ; also  $\rightsquigarrow \in C \subseteq F_C$ ; with  $\rightsquigarrow \in E$  (hypothesis),  $\rightsquigarrow \in C \cap D$ .

*Step 2 (a conclusiveness pin, sharpened by PV).* As  $C$  is conclusive and  $\rightsquigarrow \in C$ , applying conclusiveness at  $\theta = \emptyset$  gives  $\theta_0 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ,  $i^* \in I$  and  $p^* \in P$  with  $\pi_{i^*}(\uparrow \theta_0 \cap C) = \{p^*\}$ . As  $C$  is PV under  $D$  and  $\rightsquigarrow \in C \cap D$  (Step 1), corollary 5 gives  $\text{POF}_{D \setminus C}(\rightsquigarrow)$ ; applying this at  $\theta_0$  gives  $\theta_0 \sqsubseteq \theta_1 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  with  $\uparrow \theta_1 \cap D \subseteq C$ . As  $\theta_0 \sqsubseteq \theta_1$ ,  $\uparrow \theta_1 \subseteq \uparrow \theta_0$ , so  $\pi_{i^*}(\uparrow \theta_1 \cap C) \subseteq \pi_{i^*}(\uparrow \theta_0 \cap C) = \{p^*\}$ ; as  $\theta_1 \subseteq \rightsquigarrow \in C$ , this set contains  $\pi_{i^*}(\rightsquigarrow) = p^*$ , so it equals  $\{p^*\}$ . Combined with  $\uparrow \theta_1 \cap D \subseteq C$ :

$$\forall \rightsquigarrow \in \uparrow \theta_1 \cap D : \pi_{i^*}(\rightsquigarrow) = p^*. \quad (\star)$$

Let  $H := \{\alpha \mid \exists p, (i^*, \alpha, p) \in \theta_1\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  and let  $\mathbf{g} : H \rightarrow P$  be  $\mathbf{g}(\alpha) := p$  for the unique  $p$  with  $(i^*, \alpha, p) \in \theta_1$ ; as  $\theta_1 \subseteq \rightsquigarrow$ ,  $\mathbf{g} \subseteq \rightsquigarrow_{i^*}$ .

*Step 3 (propagating  $(\star)$  to every matching individual).* Fix  $i \in I$  with  $\mathbf{g} \subseteq \rightsquigarrow_i$ ; write  $g := \rightsquigarrow_i$ . We show  $p^* \in P_i$ , that is (definition 1)  $\exists a \in \mathcal{A} \mid a \hookrightarrow_i p^*$ .

If  $i = i^*$ ,  $(\star)$  at  $\rightsquigarrow := \rightsquigarrow$  directly gives  $\pi_{i^*}(\rightsquigarrow) = p^*$ , that is  $p^* \in P_{i^*}$ .

If  $i \neq i^*$ : as  $\rightsquigarrow \in E$ , the class  $J_g := \{j \in I \mid \rightsquigarrow_j = g\}$  is infinite (it contains  $i$ ), and  $\theta_1$  has finite support, so we may pick  $i'' \in J_g \setminus (\theta_1(\mathcal{A}^{<\mathbb{N}}, P) \cup \{i^*\})$ . Let  $\sigma$  be the transposition of  $i^*$  and  $i''$  (identity elsewhere) and  $\rightsquigarrow^{sw} := \rightsquigarrow \circ \sigma$ . Then  $\rightsquigarrow^{sw}(i^*) = \rightsquigarrow_{i''} = g$  and  $\mathbf{g} \subseteq g$ :  $\theta_1$ 's constraint on  $i^*$  holds in  $\rightsquigarrow^{sw}$ ; and every other  $j \in \theta_1(\mathcal{A}^{<\mathbb{N}}, P)$  satisfies  $j \notin \{i^*, i''\}$ , so  $\rightsquigarrow^{sw}(j) = \rightsquigarrow_j$ , matching  $\theta_1$  there too. Hence  $\theta_1 \subseteq \rightsquigarrow^{sw}$ . By the three invariance facts,  $\rightsquigarrow^{sw} \in E$ ,  $\neg \text{OF}_C(\rightsquigarrow^{sw})$  (as  $\neg \text{OF}_C(\rightsquigarrow)$ , Step 1), and  $\rightsquigarrow^{sw} \in F_C$  (as  $\rightsquigarrow \in F_C$ , Step 1), so  $\rightsquigarrow^{sw} \in D$ ; with  $\theta_1 \subseteq \rightsquigarrow^{sw}$ ,  $\rightsquigarrow^{sw} \in \uparrow \theta_1 \cap D$ , and  $(\star)$  gives  $\pi_{i^*}(\rightsquigarrow^{sw}) = p^*$ . As  $\pi_{i^*}$  depends only on the row  $\rightsquigarrow^{sw}(i^*) = g$  (definition 1), this is  $p^* \in P_i$ .

*Step 4 (contradiction).* Steps 2–3 give  $\forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow p^* \in P_i$ ; taking  $p := p^*$ , this is exactly the  $\mathbf{g}, p$  pattern the hypothesis excludes — contradiction.  $\square$

*Remark 15.* Taking  $F_C := P^{I \times \mathcal{A}^{<\mathbb{N}}}$  for every  $C$  (trivially anonymous, and  $C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$  always) recovers the plain statement: no such claim is PV under  $(\neg \text{OF}_C) \cap E$  alone.

As  $\text{POF}_C \Rightarrow \text{OF}_C$  (so  $\neg \text{OF}_C \subseteq \neg \text{POF}_C$ ), this is a strictly stronger conclusion than the one obtained by using  $\neg \text{POF}_C$  in place of  $\neg \text{OF}_C$  throughout: shrinking the axiom only makes it easier for a claim to be safely distinguished from the (smaller) gap  $D \setminus C$ , hence easier to be PV under it, so excluding PV under the smaller,  $\neg \text{OF}_C$ -built axiom excludes it a fortiori under the larger,  $\neg \text{POF}_C$ -built one.

$F_C$  can also be taken to be  $D_C^{\mathbf{g}', a', p'}$  (corollary 12's axiom, for any fixed  $\mathbf{g}', a', p'$  — possibly different from the  $\mathbf{g}, p$  of the hypothesis), or even  $\bigcap_k D_C^{\mathbf{g}', a_k, p_k}$  for any family of triples: both required properties are inherited automatically. Indeed,  $C \subseteq D_C^{\mathbf{g}', a', p'}$  always: for  $\rightsquigarrow \in C$ , every row  $b \in \rightsquigarrow(I)$  lies in  $S_C := \bigcup_{\rightsquigarrow'' \in C} \rightsquigarrow''(I)$  ( $\rightsquigarrow$  itself witnesses it), so  $\rightsquigarrow(I) \setminus S_C = \emptyset$  and  $D_C^{\mathbf{g}', a', p'}$ 's defining condition — every  $b \in \rightsquigarrow(I) \setminus S_C$  failing  $\mathbf{g}' \subseteq \rightsquigarrow_b \Rightarrow a' \hookrightarrow_b p'$  is distinguishable from  $S_C$  — holds vacuously. And  $D_C^{\mathbf{g}', a', p'}$  is anonymous: membership  $\rightsquigarrow \in D_C^{\mathbf{g}', a', p'}$  is a condition on the set  $\rightsquigarrow(I)$  alone (whether a row lies in  $\rightsquigarrow(I) \setminus S_C$ , fails the template, or is distinguishable from  $S_C$ , are all properties of the behavior itself, not of which individual realizes it), and  $(\rightsquigarrow \circ \sigma)(I) = \rightsquigarrow(\sigma(I)) = \rightsquigarrow(I)$  for every permutation  $\sigma$ . Both properties — containing  $C$ , being anonymous — are

preserved under arbitrary intersection (as pushforward by a bijection commutes with intersection), so the family closes under stacking finitely or infinitely many such filters together. So this whole family of claim-relative visibility filters, however combined, is powerless to rescue a claim once the underlying  $(\mathbf{g}, p)$ -pattern is absent from the world.

This breaks down for the *upfront*, non-claim-relative axiom  $D^{\mathbf{g}', a', p'} = D_{P^{\mathcal{A}^{<\mathbb{N}}, \rightsquigarrow(I)}}^{\mathbf{g}', a', p'}$  of corollary 11: it is still anonymous by the same row-set argument (with  $\rightsquigarrow(I)$ , fixed once and for all, playing the role of  $S_C$  above), but  $\rightsquigarrow \in D^{\mathbf{g}', a', p'}$  is *not* automatic from  $C$  holding — it is exactly corollary 11's nontrivial hypothesis on the true world, so it cannot be folded into a  $C \subseteq F_C$  assignment for free. Assumed explicitly instead, as an extra hypothesis  $\rightsquigarrow \in D^{\mathbf{g}', a', p'}$  alongside theorem 32's, the theorem's proof still applies verbatim (Step 1 gets  $\rightsquigarrow \in D^{\mathbf{g}', a', p'}$  from this hypothesis directly rather than from  $C \subseteq F_C$ ): so the upfront richness assumption doesn't help either, it just has to be paid for explicitly, rather than obtained for free the way the claim-relative  $D_C^{\mathbf{g}', a', p'}$  is.  $\triangle$

*Example 22.* Consider the world  $\rightsquigarrow = (J \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi\}) \cup (K \times \{\emptyset\} \times \{\varphi\}) \cup (K \times (\mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\}) \times \{\varphi'\})$  of example 2 (TODO check equality), with infinite sets of individuals  $J, K$ .

Pick some finite  $\emptyset \neq H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  such that  $1 \leq \#(\max H)$ , pick  $d \in \mathcal{A} \setminus \{a, (\max H)(1)\}$ . Define  $S = \{\emptyset \neq \alpha \mid \alpha_1 = d\} \subseteq \mathcal{A}^{<\mathbb{N}}$  as the set of paths starting with  $d$ . Let  $b_j$  be the behavior associated to  $\rightsquigarrow(J)$ , define  $b_k$  similarly. Define  $b^{\text{new}} \in P^{\mathcal{A}^{<\mathbb{N}}}$  as  $b^{\text{new}} = b_{k \upharpoonright S} \cup b_{j \upharpoonright \mathcal{A}^{<\mathbb{N}} \setminus S}$ . Consider any anonymous claim  $C \subseteq [\forall i \in I : b^{\text{new}} \upharpoonright_H \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$  that holds (for example,  $[\forall i \in I : b^{\text{new}} \upharpoonright_H \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$ ).

Observe that  $S \cap H = \emptyset$  ( $\emptyset \notin S$  by definition and  $\forall \emptyset \neq \alpha \in H : \alpha_1 = (\max H)(1) \neq d$ ) and as  $(d, a) \in S$  and  $(a, d) \notin S$ ,  $b^{\text{new}}((d, a)) = \varphi' \neq b^{\text{new}}((a, d)) = \varphi$ , thus  $\nexists p \mid a \hookrightarrow_{b^{\text{new}}} p$ . Also,  $b^{\text{new}}$  is undistinguishable from  $\rightsquigarrow(I) = \{b_j, b_k\}$ : for any  $H' \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ , since  $H'$  is a chain, either all its nonempty paths start with  $d$  — in which case  $b^{\text{new}} \upharpoonright_{H'} = b_k \upharpoonright_{H'}$  — or none do — in which case  $b^{\text{new}} \upharpoonright_{H'} = b_j \upharpoonright_{H'}$ ; and  $b^{\text{new}} \notin \rightsquigarrow(I)$ : e.g.  $b^{\text{new}}((d)) = \varphi' \neq \varphi = b_j((d))$  and  $b^{\text{new}}((a)) = \varphi \neq \varphi' = b_k((a))$ , using  $a \neq d$ . Thus, theorem 30 applies.  $\triangle$

*Remark 16.* The  $H = \emptyset$  case technically holds, but either requires to take seriously the  $\exists p$  in the claim  $C$ 's deduction ( $i_1$  has  $a$  decisive for  $\varphi$  and  $i_2$  has  $a$  decisive for  $\varphi'$ ), or if  $p$  is fixed for everybody, then the antecedent of the theorem never triggers.  $\triangle$

*Example 23.* Fix  $\varphi \neq \varphi' \in P$ ,  $a \neq b \in \mathcal{A}$ , and partition  $I = J \sqcup K$  with  $J, K$  both infinite. Define  $\rightsquigarrow$  by: for  $j \in J$ ,  $\forall \alpha : \alpha \rightsquigarrow_j \varphi \iff a \in \alpha_{\text{end}}(\alpha)$ ; for  $k \in K$ ,  $\forall \alpha : \alpha \rightsquigarrow_k \varphi \iff b \in \alpha_{\text{end}}(\alpha)$  (with  $\varphi'$  otherwise in each case). Then  $\rightsquigarrow \in E$ ;  $a \hookrightarrow_{i_j} \varphi$  and  $b \not\hookrightarrow_{i_j} \varphi$  for  $j \in J$  (as  $(b) \rightsquigarrow_j \varphi'$ );  $b \hookrightarrow_{i_k} \varphi$  and  $a \not\hookrightarrow_{i_k} \varphi$  for  $k \in K$  (as  $(a) \rightsquigarrow_k \varphi'$ ). Hence there is no  $c \in \mathcal{A}$  with  $\forall i : c \hookrightarrow_{i_i} \varphi$ . Yet  $\forall i : a \hookrightarrow_{i_i} \varphi \vee b \hookrightarrow_{i_i} \varphi$ , so  $P_i = \{\varphi\}$  for all  $i$ , and theorem 32's hypothesis fails. A suitable claim exists:  $C = [\forall i : (a) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi]$  holds in  $\rightsquigarrow$  (genuinely for  $J$ , vacuously for  $K$  since  $(a) \rightsquigarrow_k \varphi'$ ), is anonymous and conclusive, and is PV by theorem 33 (applied with  $H := \{(a)\}$ ,  $\mathbf{g} := \{((a), \varphi)\}$ ,  $c := a$ ,  $p := \varphi$ , and any  $j_0 \in J$ ).  $\triangle$

## A.9. Visibility

Given  $\mathbf{g}, a \in \mathcal{A}, p \in P, A, B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ , define  $D_{A,B}^{\mathbf{g},a,p}$  as: any behavior  $b \in A \setminus B$  failing  $\mathbf{g} \subseteq \rightsquigarrow_b \Rightarrow a \hookrightarrow_b p$  (thus satisfying  $\mathbf{g} \subseteq \rightsquigarrow_b \wedge a \not\hookrightarrow_b p$ ) is distinguishable from  $B$ .

Given  $\mathbf{g}, a \in \mathcal{A}, p \in P$ , define  $D^{\mathbf{g},a,p}$  as  $D_{P^{\mathcal{A}^{<\mathbb{N}}}, \rightsquigarrow(I)}^{\mathbf{g},a,p}$ : any behavior  $b \notin \rightsquigarrow(I)$  failing  $\mathbf{g} \subseteq \rightsquigarrow_b \Rightarrow a \hookrightarrow_b p$  is distinguishable from  $\rightsquigarrow(I)$ .

Given  $\mathbf{g}, a \in \mathcal{A}, p \in P$ , define  $D_C^{\mathbf{g},a,p}$  as  $D_{\rightsquigarrow(I), S_C}^{\mathbf{g},a,p}$ : any behavior that fails  $\mathbf{g}$  leads  $i$  to  $(a, p)$  does so visibly out of  $B$ , the types of  $C$ . (We could relax the axiom by adding  $B \not\subseteq \rightsquigarrow(I) \vee \dots$  (or  $B \subseteq \rightsquigarrow(I) \Rightarrow \dots$ ), because we need only verifiability when  $B \not\subseteq \rightsquigarrow(I)$ !)

**Theorem 33.** *Given  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  with  $\rightsquigarrow(I) \subseteq B$ , and  $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathbf{g},a,p}$  such that for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathbf{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$ , there exists an anonymous claim  $C$  that holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$ .*

*Proof.* Let  $C_0$  be the claim built in the proof of theorem 34, and set  $C := C_0 \cap [\exists i \in I \mid \mathbf{g} \subseteq \rightsquigarrow_i]$ , writing  $S_C := \bigcup_{\rightsquigarrow' \in C} \rightsquigarrow'(I)$  for the types of this (smaller) claim, and  $D := (\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$  for the axiom at stake (both evaluated at this  $C$ , as in theorem 34's proof).

*Holds, Anonymous.*  $\rightsquigarrow \in C_0$ , and the new conjunct holds at  $\rightsquigarrow$  by hypothesis, so  $\rightsquigarrow \in C$ . The new conjunct has the anonymity-friendly shape  $\exists i : Q(\rightsquigarrow(i))$ : for any permutation  $\sigma$ ,  $\exists i : \mathbf{g} \subseteq (\rightsquigarrow \circ \sigma)(i) \Leftrightarrow \exists i : \mathbf{g} \subseteq \rightsquigarrow(\sigma(i)) \Leftrightarrow \exists j : \mathbf{g} \subseteq \rightsquigarrow(j)$ , reindexing  $j := \sigma(i)$ ; combined with  $C_0$ 's anonymity,  $C \circ \sigma = C$ .

*Conclusive.* Fix  $\rightsquigarrow \in C$  and  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ . As  $\rightsquigarrow \in C$ , fix  $i^* \in I$  with  $\mathbf{g} \subseteq \rightsquigarrow_{i^*}$ , and let  $\theta' := \theta \cup (\{i^*\} \times \mathbf{g})$ , so  $\theta \sqsubseteq \theta'$  and  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$  (as in theorem 34's APTS-P paragraph, now with  $i^*, \theta$  in place of that paragraph's  $i_0, \theta_0$ ). Any  $\rightsquigarrow' \in \uparrow \theta' \cap C$  has  $\mathbf{g} \subseteq \rightsquigarrow'(i^*)$ , so  $C_0$ 's first conjunct at  $i^*$  gives  $a \hookrightarrow_{i^*} p$  in  $\rightsquigarrow'$ , hence  $\pi_{i^*}(\rightsquigarrow') = p$  by theorem 1. As  $\rightsquigarrow$  itself lies in  $\uparrow \theta' \cap C$ , this set is nonempty, so  $\pi_{i^*}(\uparrow \theta' \cap C) = \{p\}$ :  $(\theta', i^*)$  witnesses conclusiveness at  $(\rightsquigarrow, \theta)$ . As both were arbitrary,  $C$  is conclusive.

*PV under  $D$ .* By corollary 5, it suffices to show  $\text{POF}_{D \setminus C}(\rightsquigarrow)$  for every  $\rightsquigarrow \in C \cap D$ .

$D \subseteq C_0$ . Steps 1, A and B of theorem 34's proof only ever use  $C_0$ 's first and second conjuncts (never a bound on  $C$  from above), so they apply verbatim to  $C$  in place of that proof's own claim, with this  $C$ 's own  $S_C$ : any  $\rightsquigarrow' \in E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$  outside  $C_0$  satisfies  $\text{POF}_C(\rightsquigarrow')$ , that is  $\rightsquigarrow' \notin \neg \text{POF}_C$ . So no  $\rightsquigarrow' \in D$  lies outside  $C_0$ :  $D \subseteq C_0$ .

Fix  $\rightsquigarrow \in C \cap D$  and  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ ; reuse  $i^*, \theta'$  from the *Conclusive* paragraph above. Suppose  $\rightsquigarrow'' \in D$  extends  $\theta'$ . By  $D \subseteq C_0$ ,  $\rightsquigarrow'' \in C_0$ ; and  $\theta' \subseteq \rightsquigarrow''$  gives  $\mathbf{g} \subseteq \rightsquigarrow''(i^*)$ , so  $\rightsquigarrow''$  also satisfies  $C$ 's extra conjunct (witnessed by  $i^*$ ), hence  $\rightsquigarrow'' \in C$ . So no  $\rightsquigarrow'' \in D \setminus C$  extends  $\theta'$ , that is,  $\theta' \not\subseteq (D \setminus C)^\subseteq$ . As  $\theta$  was arbitrary,  $\text{POF}_{D \setminus C}(\rightsquigarrow)$  holds; as  $\rightsquigarrow \in C \cap D$  was arbitrary,  $C$  satisfies PV under  $D$ .  $\square$

**Conjecture 2.** *Given  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  with  $\rightsquigarrow(I) \subseteq B$  such that for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathbf{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$ , if there exists an anonymous claim  $C$  that holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E$ , then  $\rightsquigarrow \in D^{\mathbf{g},a,p}$ .*

*Disproof.* We build a single world satisfying the hypotheses for a fixed  $H, \mathbf{g}, a, p$ , together with a witnessing claim  $C$ , yet with  $\rightsquigarrow \notin D^{\mathbf{g}, a, p}$ : nothing ties the witness  $C$  furnished by the hypothesis to the specific pattern named in the conclusion, so it can be supplied by an entirely different, unrelated part of the world.

Fix  $a \neq c \in \mathcal{A}$  and partition  $I$  into infinite  $J, K$ , with  $\forall i \in J, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$ , and  $\forall i \in K, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$  if  $c \in \alpha_{\text{end}}(\alpha)$ ,  $\alpha \rightsquigarrow_i \varphi'$  otherwise (so  $(a) \rightsquigarrow_i \varphi'$  throughout  $K$ , as  $c \neq a$ ); let  $b_j, b_k$  be the behaviors associated to  $\rightsquigarrow(J), \rightsquigarrow(K)$ . Take  $H := \{(a)\}$  (so  $\#(\max H) = 1$ ),  $\mathbf{g} := \{((a), \varphi)\}$ ,  $p := \varphi$ : the pattern  $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$  holds throughout  $J \cup K$  (trivially at  $J$ , vacuously at  $K$  since  $b_k((a)) = \varphi' \neq \varphi$ ) and fires at any  $i \in J$ .

$\rightsquigarrow \notin D^{\mathbf{g}, a, p}$  on  $J \cup K$ . Pick  $d \in \mathcal{A} \setminus \{a, c\}$ ,  $S := \{\alpha \neq \emptyset \mid \alpha_1 = d\}$ , and  $g := b_k|_S \cup b_j|_{\mathcal{A}^{<\mathbb{N}} \setminus S}$ . Any  $H' \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  is a chain, hence lies entirely inside  $S$  or entirely outside it, so  $g$  matches  $b_k$  or  $b_j$  on  $H'$ :  $g$  is undistinguishable from  $\{b_j, b_k\}$ . As  $d \neq a$ ,  $(a) \notin S$ , so  $g((a)) = b_j((a)) = \varphi$ :  $g$  matches  $\mathbf{g}$ . As  $(d, a) \in S$  and  $c \notin \{d, a\}$ ,  $g((d, a)) = b_k((d, a)) = \varphi'$ , and  $a \in \alpha_{\text{end}}((d, a))$ :  $a$  is not decisive for  $\varphi$  in  $g$ . Yet  $g \notin \{b_j, b_k\}$  ( $g((d)) = \varphi' \neq \varphi = b_j((d))$ ;  $g((a)) = \varphi \neq \varphi' = b_k((a))$ ):  $g$  is exactly the counterexample  $D^{\mathbf{g}, a, p}$  forbids.

Adjoin a third infinite set  $L$ , disjoint from  $J \cup K$ , with  $\rightsquigarrow_i := b_l$  constant equal to some fixed  $\psi \in P \setminus \{\varphi, \varphi'\}$  for every  $i \in L$ ; write  $I := J \cup K \cup L$ , keeping  $\rightsquigarrow$  unchanged on  $J \cup K$ . As  $J, K, L$  are infinite and pairwise disjoint and  $b_j, b_k, b_l$  pairwise distinct,  $\rightsquigarrow \in E$ . The pattern above still holds on the enlarged  $I$ :  $\mathbf{g}$ 's only value is  $\varphi \neq \psi = b_l(\alpha)$  for every  $\alpha$ , so  $\neg(\mathbf{g} \subseteq b_l)$ , and the antecedent fails vacuously on  $L$ . And  $g$  remains a witness of  $\rightsquigarrow \notin D^{\mathbf{g}, a, p}$ : it is still undistinguishable from  $\rightsquigarrow(I) = \{b_j, b_k, b_l\}$  (it already matched  $b_j$  or  $b_k$  on every  $H' \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ), it still lies outside  $\rightsquigarrow(I)$  (its values lie in  $\{\varphi, \varphi'\}$ , so  $g \neq b_l$ ), and  $a$  is still not decisive for  $\varphi$  in  $g$ .

*A witness, entirely on  $L$ .* Let  $H_1 := \{\emptyset, (a)\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ,  $\mathbf{g}_1 : H_1 \rightarrow P$  constant  $\psi$ , and  $p_1 := \psi$  (reusing the same action  $a$  — nothing below needs a fresh one). At  $i \in L$ :  $\mathbf{g}_1 \subseteq b_l$  (as  $b_l$  is constant  $\psi$ ) and  $a$  is (trivially) decisive for  $\psi$  in the constant row  $b_l$ : the pattern holds and fires on  $L$ . It holds vacuously on  $J \cup K$ :  $b_j(\emptyset) = \varphi \neq \psi$ , and  $b_k(\emptyset) = \varphi' \neq \psi$  (as  $c \notin \alpha_{\text{end}}(\emptyset)$ ), so  $\neg(\mathbf{g}_1 \subseteq b_j)$  and  $\neg(\mathbf{g}_1 \subseteq b_k)$ .

$\rightsquigarrow \in D^{\mathbf{g}_1, a, p_1}$ . Let  $b \notin \rightsquigarrow(I) = \{b_j, b_k, b_l\}$  satisfy  $\mathbf{g}_1 \subseteq b$  (so  $b(\emptyset) = b((a)) = \psi$ ) and  $a \not\hookrightarrow_b p_1$ , witnessed by some  $\beta$  with  $a \in \alpha_{\text{end}}(\beta)$  (so  $\beta \neq \emptyset$ ) and  $b(\beta) \neq \psi$ . At the chain  $\{\emptyset, \beta\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ,  $b|_{\{\emptyset, \beta\}} = (\psi, b(\beta))$  with  $b(\beta) \neq \psi$ , while  $b_j|_{\{\emptyset, \beta\}} = (\varphi, \varphi)$ ,  $b_k|_{\{\emptyset, \beta\}} = (\varphi', b_k(\beta))$ , and  $b_l|_{\{\emptyset, \beta\}} = (\psi, \psi)$ : the first coordinate  $\psi \notin \{\varphi, \varphi'\}$  already rules out  $b_j$  and  $b_k$ , and the second coordinate  $b(\beta) \neq \psi$  rules out  $b_l$ . So  $b$  is distinguishable from  $\rightsquigarrow(I)$ :  $\rightsquigarrow \in D^{\mathbf{g}_1, a, p_1}$ .

By corollary 11, applied with this  $H_1, \mathbf{g}_1, a, p_1$ , some anonymous claim  $C$  holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E$ .

Taking  $B := P^{\mathcal{A}^{<\mathbb{N}}}$  (which trivially satisfies  $\rightsquigarrow(I) \subseteq B$  and, since it appears nowhere else in the statement, plays no further role), the conjecture's hypotheses hold at this  $\rightsquigarrow$  for  $H, \mathbf{g}, a, p$  as above, and  $C$  witnesses its antecedent — yet  $\rightsquigarrow \notin D^{\mathbf{g}, a, p}$ : the conjecture is false. The obstruction (on  $J \cup K$ ) and the witness (on  $L$ ) sit on disjoint, mutually irrelevant parts of  $I$ , precisely because nothing in the antecedent ties the claim  $C$  it

furnishes to the specific  $\mathbf{g}, a, p$  named in the conclusion.  $\square$

**Theorem 34.** *Given  $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$  with  $\rightsquigarrow(I) \subseteq B$ , and  $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathbf{g}, a, p}$  such that for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathbf{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$ , there exists an anonymous claim  $C$  that holds, is APTS and provable by  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g}, a, p}$ .*

*Proof.* Write  $\text{Bad} := \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \mid b \notin B \wedge \mathbf{g} \subseteq b \wedge a \not\hookrightarrow_b p\}$ . As  $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathbf{g}, a, p}$ , fix, for each  $b \in \text{Bad}$ , some  $H_b \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  with  $b|_{H_b} \notin B|_{H_b}$ . Take

$$C := [\forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\forall i \in I, b \in \text{Bad} : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}].$$

*Holds.* The first conjunct holds at  $\rightsquigarrow$ : this is exactly the hypothesis  $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ . For the second, fix  $b \in \text{Bad}$ : as  $\rightsquigarrow(I) \subseteq B$ , every  $\rightsquigarrow_i \in B$ , so  $b|_{H_b} \notin B|_{H_b}$  gives  $\forall i \in I : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}$ .

*Anonymous.* Write  $Q_1(f) := [\mathbf{g} \subseteq f \Rightarrow a \hookrightarrow_f p]$  and  $Q_2(f) := [\forall b \in \text{Bad} : f|_{H_b} \neq b|_{H_b}]$ , properties of a single row  $f$  not depending on  $i$  (Bad and each  $H_b$  are fixed once and for all), so that  $C = [\forall i : Q_1(\rightsquigarrow(i)) \wedge Q_2(\rightsquigarrow(i))]$ . Fix a permutation  $\sigma$  of  $I$  and  $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ . As  $(\rightsquigarrow \circ \sigma)(i) = \rightsquigarrow(\sigma(i))$ ,

$$\rightsquigarrow \circ \sigma \in C \Leftrightarrow \forall i \in I : Q_1(\rightsquigarrow(\sigma(i))) \wedge Q_2(\rightsquigarrow(\sigma(i))) \Leftrightarrow \forall j \in I : Q_1(\rightsquigarrow(j)) \wedge Q_2(\rightsquigarrow(j)) \Leftrightarrow \rightsquigarrow \in C,$$

reindexing  $j := \sigma(i)$ , a bijection of  $I$ . As  $\rightsquigarrow, \sigma$  were arbitrary,  $C \circ \sigma = C$ :  $C$  is anonymous.

*APTS-P (hence APTS).* Fix  $i_0 \in I$  with  $\theta_0 := \{i_0\} \times \mathbf{g} \subseteq \rightsquigarrow$  (hypothesis). For any  $\rightsquigarrow' \in \uparrow \theta_0 \cap C$ :  $\theta_0 \subseteq \rightsquigarrow'$  gives  $\mathbf{g} \subseteq \rightsquigarrow'(i_0)$ , so  $C$ 's first conjunct at  $i_0$  gives  $a \hookrightarrow_{i_0} p$  in  $\rightsquigarrow'$  directly, that is  $p \in P_{i_0}(\rightsquigarrow')$ ; by theorem 1,  $P_{i_0}(\rightsquigarrow') = \{p\}$  exactly, that is  $\pi_{i_0}(\rightsquigarrow') = p$ . Since moreover  $\rightsquigarrow \in \uparrow \theta_0 \cap C$  (as  $\theta_0 \subseteq \rightsquigarrow$  and  $\rightsquigarrow \in C$ ), this set is nonempty, so  $\pi_{i_0}(\uparrow \theta_0 \cap C) = \{p\}$ : as  $p \in P$  and  $\theta_0$  is feasible, this is APTS-P, witnessed by  $(\theta_0, i_0, p)$ ; in particular  $C$  is APTS.

*Provable by  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g}, a, p}$ .* Write  $S_C := \bigcup_{\rightsquigarrow' \in C} \rightsquigarrow'(I)$  for the types of  $C$ , evaluated at the claim  $C$  just constructed above (not circular:  $S_C$ , and the axiom  $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g}, a, p}$ , are only ever evaluated at this  $C$ , never used to construct it).

*Step 1 ( $S_C$ -types are decisive).* For  $h \in S_C$ , some  $\rightsquigarrow'' \in C$  and  $i'' \in I$  satisfy  $\rightsquigarrow''(i'') = h$ ;  $C$ 's first conjunct at  $i''$  gives  $\mathbf{g} \subseteq h \Rightarrow a \hookrightarrow_h p$ . Contrapositively, any  $h$  with  $\mathbf{g} \subseteq h \wedge a \not\hookrightarrow_h p$  satisfies  $h \notin S_C$ .

We show  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g}, a, p} \subseteq C$ . Fix  $\rightsquigarrow' \in E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g}, a, p}$  with  $\rightsquigarrow' \notin C$ .

*Case B: the second conjunct fails* (the template holds, but its extra content — avoiding collision with Bad — does not). Some  $i \in I$ ,  $b \in \text{Bad}$  satisfy  $\rightsquigarrow'(i)|_{H_b} = b|_{H_b}$ ; write  $g := \rightsquigarrow'(i)$ , so  $g|_{H_b} = b|_{H_b}$ .

Fix any  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$ . As  $\rightsquigarrow' \in E$ ,  $J_g := \{j \in I \mid \rightsquigarrow'(j) = g\}$  is infinite; as  $\theta$  is finite,  $\theta(\mathcal{A}^{<\mathbb{N}}, P)$  is finite, so  $J_g \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \neq \emptyset$ . Pick  $i'$  in this set and let  $\theta_b^{i'} := \{(i', \alpha, g(\alpha)) \mid \alpha \in H_b\}$ . As  $H_b \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ,  $\theta_b^{i'} \in \Theta$ , and  $\theta_b^{i'} \subseteq \rightsquigarrow'$  (as  $\rightsquigarrow'(i') = g$ ).

If some  $\rightsquigarrow''' \in C$  extended  $\theta_b^{i'}$ , then  $f := \rightsquigarrow'''(i')$  would satisfy  $f|_{H_b} = g|_{H_b} = b|_{H_b}$ , contradicting  $C$ 's second conjunct at  $i', b$ , applied to  $\rightsquigarrow'''$ . So  $\theta_b^{i'} \notin C^\subseteq$ .

Let  $\theta' := \theta \cup \theta_b^{i'}$ . Since  $i'$  is untouched by  $\theta$ ,  $\theta \sqsubseteq \theta'$ , and  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow')$ . Since  $\theta_b^{i'} \subseteq \theta'$  and  $C^\subseteq$  is downward closed,  $\theta' \in C^\subseteq$  would force  $\theta_b^{i'} \in C^\subseteq$ , contradicting the above; so  $\theta' \notin C^\subseteq$ . As  $\theta$  was arbitrary (the same  $b, H_b$  serve throughout; only the clone  $i'$  depends on  $\theta$ ),  $\text{POF}_C(\rightsquigarrow')$  holds.

*Case A: the second conjunct holds but the first fails* (falsifying the template itself). Some  $i \in I$  has  $g := \rightsquigarrow'(i)$  with  $\mathfrak{g} \subseteq g$  and  $a \not\rightarrow_g p$ ; by Step 1,  $g \notin S_C$ . Moreover  $g \notin \text{Bad}$ : otherwise the second conjunct, evaluated at this  $i$  and  $b := g \in \text{Bad}$ , would require  $g_{\upharpoonright H_g} \neq g_{\upharpoonright H_g}$ , absurd. Since  $g$  satisfies  $\mathfrak{g} \subseteq g \wedge a \not\rightarrow_g p$  yet  $g \notin \text{Bad} = \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus B \mid \mathfrak{g} \subseteq b \wedge a \not\rightarrow_b p\}$ , this forces  $g \in B$ . So  $g \in (\rightsquigarrow'(I) \cap B) \setminus S_C$ , and as  $\rightsquigarrow' \in D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$ , this yields some  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  with  $g_{\upharpoonright H} \notin S_{C \upharpoonright H}$ .

Fix any  $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$ . As  $\rightsquigarrow' \in E$ ,  $J_g := \{j \in I \mid \rightsquigarrow'(j) = g\}$  is infinite; as  $\theta$  is finite,  $\theta(\mathcal{A}^{<\mathbb{N}}, P)$  is finite, so  $J_g \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \neq \emptyset$ . Pick  $i'$  in this set and let  $\theta_H^{i'} := \{(i', \alpha, g(\alpha)) \mid \alpha \in H\}$ . As  $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ,  $\theta_H^{i'} \in \Theta$ , and  $\theta_H^{i'} \subseteq \rightsquigarrow'$  (as  $\rightsquigarrow'(i') = g$ ).

If some  $\rightsquigarrow''' \in C$  extended  $\theta_H^{i'}$ , then  $f := \rightsquigarrow'''(i')$  would satisfy  $f_{\upharpoonright H} = g_{\upharpoonright H}$  while  $f \in \rightsquigarrow'''(I) \subseteq S_C$ , giving  $g_{\upharpoonright H} \in S_{C \upharpoonright H}$  — contradiction. So  $\theta_H^{i'} \notin C^\subseteq$ .

Let  $\theta' := \theta \cup \theta_H^{i'}$ . Since  $i'$  is untouched by  $\theta$ ,  $\theta \sqsubseteq \theta'$ , and  $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow')$ . Since  $\theta_H^{i'} \subseteq \theta'$  and  $C^\subseteq$  is downward closed,  $\theta' \in C^\subseteq$  would force  $\theta_H^{i'} \in C^\subseteq$ , contradicting the above; so  $\theta' \notin C^\subseteq$ . As  $\theta$  was arbitrary (the same  $g, H$  serve throughout; only the clone  $i'$  depends on  $\theta$ ),  $\text{POF}_C(\rightsquigarrow')$  holds.

Either way,  $\text{POF}_C(\rightsquigarrow')$  holds, so  $\rightsquigarrow' \notin \neg \text{POF}_C$ . As  $\rightsquigarrow' \in E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} \setminus C$  was arbitrary,  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} \subseteq C$ .  $\square$

*Remark 17.* Taking  $B = \rightsquigarrow(I)$  trivializes the filter axiom:  $\rightsquigarrow(I) \subseteq S_C$  (every row of  $\rightsquigarrow$  is, by *Holds*, a type of  $C$ ), so  $\rightsquigarrow(I) \cap B \subseteq S_C$  always, the antecedent  $(\rightsquigarrow(I) \cap B) \setminus S_C$  is never met, and  $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ . Taking  $B = P^{\mathcal{A}^{<\mathbb{N}}}$  instead makes  $\text{Bad} = \emptyset$ , so  $C_0$ 's second conjunct vanishes and  $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} = D_{\rightsquigarrow(I), S_C}^{\mathfrak{g}, a, p} = D_C^{\mathfrak{g}, a, p}$ , the full axiom of corollary 12. This only ever used *Holds* and the definition of *Bad*, so it holds verbatim for theorem 33's construction too: either theorem, at  $B = \rightsquigarrow(I)$ , reduces to corollary 11, and at  $B = P^{\mathcal{A}^{<\mathbb{N}}}$ , to corollary 12. Between these extremes, the stronger the upfront axiom on  $\rightsquigarrow$  (the smaller  $B$ ), the weaker the residual axiom needed in the filter (the smaller  $\rightsquigarrow(I) \cap B$  to test) — and conversely.  $\triangle$

**Corollary 11.** *Given  $\rightsquigarrow \in D^{\mathfrak{g}, a, p}$  such that for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathfrak{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathfrak{g} \subseteq \rightsquigarrow_i$ , there exists an anonymous claim that holds, is conclusive and PV under  $\neg \text{POF}_C \cap E$ .*

*Proof.* Apply theorem 33 with  $B := \rightsquigarrow(I)$ : the hypothesis  $\rightsquigarrow(I) \subseteq B$  is trivial, and  $D_{P^{\mathcal{A}^{<\mathbb{N}}}, \rightsquigarrow(I)}^{\mathfrak{g}, a, p} = D^{\mathfrak{g}, a, p}$  is exactly the hypothesis here. By the remark following theorem 34, at this  $B$  the filter axiom  $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$  trivializes to  $P^{I \times \mathcal{A}^{<\mathbb{N}}}$ , so PV under  $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$  is exactly PV under  $\neg \text{POF}_C \cap E$ .  $\square$

**Corollary 12.** *Given, for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathfrak{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathfrak{g} \subseteq \rightsquigarrow_i$ , there exists an anonymous claim  $C$  that holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E \cap D_C^{\mathfrak{g}, a, p}$ .*

*Proof.* Apply theorem 33 with  $B := P^{\mathcal{A}^{<\mathbb{N}}}$ :  $\rightsquigarrow(I) \subseteq B$  trivially, and  $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, P^{\mathcal{A}^{<\mathbb{N}}}}^{\mathfrak{g}, a, p}$  holds vacuously, its antecedent  $P^{\mathcal{A}^{<\mathbb{N}}} \setminus B = \emptyset$  being empty — no distinguishability hypothesis on  $\rightsquigarrow$  is needed at all. At this  $B$ ,  $\text{Bad} = \emptyset$ , so  $C_0$ 's second conjunct vanishes, leaving  $C = [\forall i \in I : \mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\exists i \in I \mid \mathfrak{g} \subseteq \rightsquigarrow_i]$ ; and  $\rightsquigarrow(I) \cap B = \rightsquigarrow(I)$ , so the filter axiom  $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$  is the full  $D_{\rightsquigarrow(I), S_C}^{\mathfrak{g}, a, p} = D_C^{\mathfrak{g}, a, p}$ . (All of the distinguishing burden is now paid at each candidate world rather than upfront on  $\rightsquigarrow$ , in contrast to corollary 11.)  $\square$

**Corollary 13.** *Given, for some  $H$  with  $1 \leq \#(\max H)$ ,  $\mathfrak{g} : H \rightarrow P$ ,  $a, p$ ,  $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$ , and  $\exists i \mid \mathfrak{g} \subseteq \rightsquigarrow_i$ , there exists an anonymous claim  $C$  that holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E \cap D$ , where  $D$  is*

$$\left[ \forall g \in \rightsquigarrow(I) \text{ with } \mathfrak{g} \subseteq g \wedge a \not\hookrightarrow_g p, \right. \\ \left. \exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \mathfrak{g}^{-1}(P) \cup \{\alpha\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) \wedge a \in \alpha_{\text{end}}(\alpha) \wedge \alpha \not\hookrightarrow_g p \right].$$

*Proof.* Let  $C := [\forall i \in I : \mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\exists i \in I \mid \mathfrak{g} \subseteq \rightsquigarrow_i]$ , the claim of corollary 12, which holds, is conclusive and PV under  $(\neg \text{POF}_C) \cap E \cap D_C^{\mathfrak{g}, a, p}$ ; it remains to show  $D_C^{\mathfrak{g}, a, p} = D$ .

As  $\text{Bad} = \emptyset$  at this  $C$  (corollary 12's proof),  $S_C = \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \mid \mathfrak{g} \subseteq b \Rightarrow a \hookrightarrow_b p\}$ . A row  $g$  failing  $\mathfrak{g} \subseteq g \Rightarrow a \hookrightarrow_g p$  never lies in  $S_C$ , so the condition " $g \in \rightsquigarrow(I) \setminus S_C$ " in the definition of  $D_C^{\mathfrak{g}, a, p}$  is automatic whenever  $\mathfrak{g} \subseteq g \wedge a \not\hookrightarrow_g p$ ; unfolding distinguishability,  $\rightsquigarrow \in D_C^{\mathfrak{g}, a, p}$  says exactly: for every  $g \in \rightsquigarrow(I)$  with  $\mathfrak{g} \subseteq g \wedge a \not\hookrightarrow_g p$ , some  $H_0 \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$  has no  $b \in S_C$  with  $b|_{H_0} = g|_{H_0}$ . Fix such a  $g$ ; we show this holds iff some  $\alpha$  as in  $D$ 's definition exists.

*If such an  $\alpha$  exists,* take  $H_0 := \mathfrak{g}^{-1}(P) \cup \{\alpha\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ . If some  $b \in S_C$  matched  $g$  on  $H_0$ , then matching  $g$  on  $\mathfrak{g}^{-1}(P) \subseteq H_0$  would give  $\mathfrak{g} \subseteq b$  (as  $\mathfrak{g} \subseteq g$ ), hence  $a \hookrightarrow_b p$  by  $b \in S_C$ , hence  $b(\alpha) = p$  (as  $a \in \alpha_{\text{end}}(\alpha)$ ); but matching  $g$  at  $\alpha \in H_0$  gives  $b(\alpha) = g(\alpha) \neq p$  — contradiction. So no such  $b$  exists:  $H_0$  witnesses  $g$  distinguishable from  $S_C$ .

*If no such  $\alpha$  exists,* fix any  $H_0 \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ ; we build  $b \in S_C$  matching  $g$  on  $H_0$ . If  $\mathfrak{g}^{-1}(P) \not\subseteq H_0$ , pick  $\gamma \in \mathfrak{g}^{-1}(P) \setminus H_0$  and  $q \in P \setminus \{\mathfrak{g}(\gamma)\}$  ( $\#P \geq 2$ ); the row  $b$  with  $b(\gamma) = q$ ,  $b = g$  on  $H_0$  (consistent, as  $\gamma \notin H_0$ ), completed arbitrarily elsewhere, fails  $\mathfrak{g} \subseteq b$ , so satisfies the template vacuously:  $b \in S_C$ . Otherwise  $\mathfrak{g}^{-1}(P) \subseteq H_0$ ; since  $H_0$  is a chain, every  $\beta \in H_0$  with  $a \in \alpha_{\text{end}}(\beta)$  is chain-compatible with  $\mathfrak{g}^{-1}(P) \subseteq H_0$ , so, by hypothesis (no valid  $\alpha$ ), every such  $\beta$  has  $g(\beta) = p$ . The row  $b$  with  $b = g$  on  $H_0$  and  $b(\beta) = p$  for every  $\beta$  with  $a \in \alpha_{\text{end}}(\beta)$  outside  $H_0$ , completed arbitrarily on the rest, then satisfies  $\mathfrak{g} \subseteq b$  (as  $\mathfrak{g}^{-1}(P) \subseteq H_0$ ) and  $a \hookrightarrow_b p$  (using the previous sentence for the  $a$ -ending points inside  $H_0$ ), so  $b \in S_C$ . Either way, some  $b \in S_C$  matches  $g$  on  $H_0$ ; as  $H_0$  was arbitrary,  $g$  is not distinguishable from  $S_C$ .  $\square$

*Remark 18.* Compare to corollary 12: writing  $C' := [\forall i : Q_1(\rightsquigarrow(i))] \cap [\exists i \in I \mid \mathfrak{g} \subseteq \rightsquigarrow_i]$  (corollary 12's own claim, for the same  $\mathfrak{g}, a, p$ ;  $Q_1$  and  $\text{Bad}$  as in theorem 34's proof, instantiated at  $B = \rightsquigarrow(I)$ ),  $C = C' \cap [\forall i, b \in \text{Bad} : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}] \subseteq C'$  (the extra  $\text{Bad}$ -avoidance conjunct commutes past the shared witness conjunct): the claim proved

here is stronger (strictly so unless  $\text{Bad} = \emptyset$ ), as it additionally excludes, on each realized row, collision with any  $\mathbf{g}$ -matching, decisiveness-failing non-real behavior on its own witnessing window, while  $C'$  only demands decisiveness on  $\mathbf{g}$ -matching rows.

$D^{\mathbf{g},a,p}$  and  $D_{C'}^{\mathbf{g},a,p}$  are not comparable by inclusion:  $D^{\mathbf{g},a,p}$  quantifies over behaviors  $b \notin \rightsquigarrow(I)$ , tested against  $\rightsquigarrow(I)$  itself, whereas  $D_{C'}^{\mathbf{g},a,p}$  quantifies over behaviors  $h \in \rightsquigarrow(I) \setminus S_{C'}$ , tested against the external set  $S_{C'}$  (the types of  $C'$ ).

Both proofs move a witness to an untouched clone using  $E$  in the same way, but differ in when the distinguishing axiom is cashed in. Here,  $\neg \text{POF}_C \cap E \subseteq C$  outright:  $\rightsquigarrow \in D^{\mathbf{g},a,p}$  is a hypothesis on the true world alone, spent only once, upfront, to fix  $B$  and the  $H_b$ 's, before any candidate world is examined, so it drops out of corollary 11's conclusion entirely. In corollary 12, by contrast,  $\rightsquigarrow \in D_{C'}^{\mathbf{g},a,p}$  must instead be invoked afresh at each tested  $\rightsquigarrow$ , since its witnessing window depends on that world's own failing row  $g := \rightsquigarrow(i)$ , which varies with  $\rightsquigarrow$ ; so that axiom cannot similarly be dropped from the conclusion. So the asymmetry is not about  $E$ , but about when the distinguishing axiom is cashed in: once, from  $\rightsquigarrow$  alone, for the stronger,  $\rightsquigarrow$ -dependent claim  $C$ ; or per candidate world, for the weaker,  $\rightsquigarrow$ -independent claim  $C'$ .

The same two instantiations ( $B := \rightsquigarrow(I)$  and  $B := P^{st < \mathbb{N}}$ ) apply just as well to theorem 34 itself, giving analogues of corollaries 11 and 12 for the original, unaugmented claim  $C_0$ : an anonymous, holding claim that is APTS and provable by  $\neg \text{POF}_C \cap E$ , respectively APTS and provable by  $(\neg \text{POF}_C) \cap E \cap D_{C_0}^{\mathbf{g},a,p}$ .  $\triangle$